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# VIOLENCE, INSTITUTIONS, AND ECONOMIC GROWTH

Evidencia de Panel para Centroamérica, Colombia y República Dominicana

|                   |                                    |
|-------------------|------------------------------------|
| Autor             | Pipeline Económico — Auto-generado |
| Fecha             | 03 July 2026                       |
| Observaciones     | 179                                |
| Países / Unidades | 8                                  |
| Período temporal  | 2000–2024                          |
| Script maestro    | run_pipeline.py                    |
| Versión pipeline  | 2.0                                |

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# 1. Resumen Ejecutivo

Este informe presenta los resultados del pipeline econométrico que estima el efecto causal de la violencia (medida por la tasa de homicidios) sobre las instituciones, la inversión extranjera directa y el crecimiento económico en un panel de países de Centroamérica, Colombia y República Dominicana para el período 2000–2024. La estimación principal emplea efectos fijos bidireccionales (país × año) con errores estándar clusterizados a nivel de país.

|                                  |         |
|----------------------------------|---------|
| Número de países (N)             | 8       |
| Períodos distintos (T)           | 24      |
| Observaciones totales            | 179     |
| Panel balanceado                 | No      |
| Varianza explicada PC1 (inst.)   | 0.789   |
| Alfa de Cronbach (inst.)         | 0.863   |
| EQ1: $\beta$ (violencia → inst.) | -0.1404 |
| EQ1: p-valor (CL)                | 0.3032  |
| EQ2: $\beta$ (inst. → FDI)       | 0.9000  |
| EQ2: p-valor (CL)                | 0.0548  |
| EQ3: $\beta$ (FDI → crecimiento) | 0.1371  |
| EQ3: p-valor FDI (CL)            | 0.2126  |

## ■ Lectura de resultados

El coeficiente de EQ1 es negativo, indicando que mayor violencia se asocia con instituciones más débiles, aunque el p-valor sugiere que no se rechaza  $H_0$  con los umbrales convencionales bajo este esquema de inferencia. EQ2 muestra que mejores instituciones atraen más FDI ( $p \approx 0.055$ , marginalmente significativo). El mecanismo completo (violencia → inst. → FDI → crecimiento) requiere interpretación conjunta de los tres modelos y los resultados bootstrap.

## 2. Información del Dataset

El panel cubre 8 países de América Central, Colombia y República Dominicana. Los datos provienen del Banco Mundial (WDI, WGI) y fuentes nacionales de estadísticas de crimen.

| COL | Colombia        |
|-----|-----------------|
| CRI | Costa Rica      |
| DOM | Rep. Dominicana |
| GTM | Guatemala       |
| HND | Honduras        |
| NIC | Nicaragua       |
| PAN | Panamá          |
| SLV | El Salvador     |

Período temporal: 2000–2024. Número de períodos distintos observados: 24. El panel no es balanceado.

### Varianza within-país

La varianza within-país indica cuánta variación temporal existe dentro de cada unidad. Valores altos justifican el uso de efectos fijos.

| homicide_rate_log  | 53.68% |
|--------------------|--------|
| inst_avg           | 24.98% |
| inst_pca           | 24.54% |
| fdi_percent_gdp    | 74.44% |
| gdp_growth         | 96.20% |
| gdp_per_capita_log | 52.45% |

### 3. Variables Utilizadas

A continuación se listan las variables dependientes, independientes clave y controles utilizados en el pipeline, junto con su transformación aplicada.

|                     |                                  |                              |                              |
|---------------------|----------------------------------|------------------------------|------------------------------|
| homicide_rate_log   | Tasa de homicidios por 100K hab. | log(x+1), lag 1 año          | Independiente clave (EQ1)    |
| inst_avg / inst_pca | Índice institucional compuesto   | Promedio estandarizado / PC1 | Mediador (EQ1→EQ2)           |
| rule_of_law         | Estado de derecho (WGI)          | Sin transformar (−2.5 a 2.5) | Componente inst.             |
| control_corruption  | Control de corrupción (WGI)      | Sin transformar (−2.5 a 2.5) | Componente inst.             |
| political_stability | Estabilidad política (WGI)       | Sin transformar (−2.5 a 2.5) | Componente inst.             |
| fdi_percent_gdp     | IED como % del PIB               | Sin transformar              | Dependiente EQ2 / Indep. EQ3 |
| gdp_growth          | Crecimiento real del PIB (%)     | Sin transformar              | Dependiente EQ3              |
| gdp_per_capita_log  | PIB per cápita (USD cte.)        | log(x)                       | Control                      |
| inflation           | Inflación anual (%)              | Sin transformar              | Control                      |
| exports_percent_gdp | Exportaciones como % del PIB     | Sin transformar              | Control                      |
| population_log      | Población total                  | log(x)                       | Control                      |
| unemployment        | Tasa de desempleo (%)            | Sin transformar              | Control                      |

#### Índice Institucional — PCA

Se extrajo el primer componente principal (PC1) de las tres dimensiones WGI como medida sintética de calidad institucional. PC1 explica el 0.8% de la varianza. El alfa de Cronbach es 0.863 (KMO = 0.534).

|                     |        |
|---------------------|--------|
| rule_of_law         | 0.6300 |
| control_corruption  | 0.5865 |
| political_stability | 0.5091 |

#### ■ Validez del índice institucional

Un alfa de Cronbach > 0.80 indica alta consistencia interna de las tres dimensiones. El primer componente explica 0.8% de la varianza, justificando la reducción dimensional. Todos los loadings son positivos, confirmando que PC1 captura 'calidad institucional general' y no una dimensión específica.

## 4. Estadísticos Descriptivos

Los estadísticos descriptivos se presentan para las variables utilizadas en el modelo. Las transformaciones logarítmicas reducen la asimetría de las distribuciones originales.

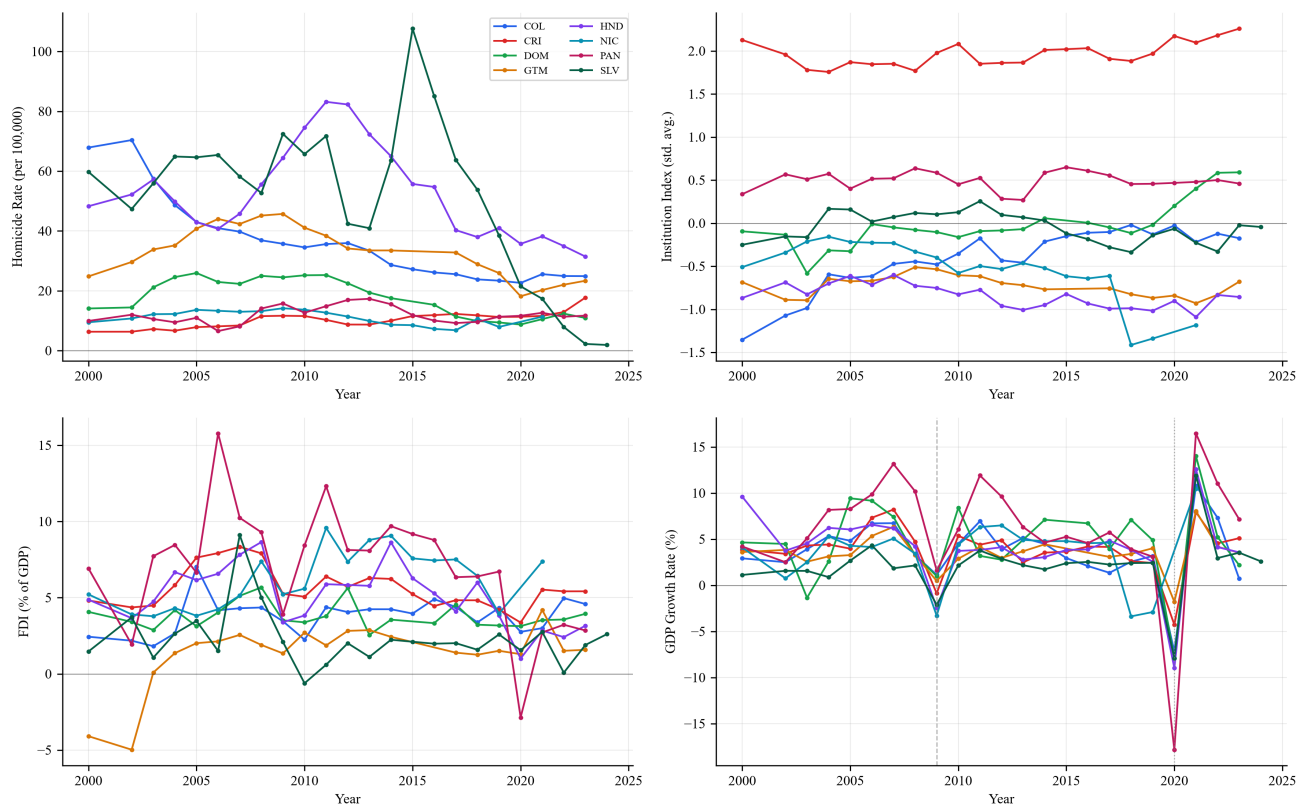
| homicide_rate_log   | 179.0 | 3.139  | 0.698  | 1.86    | 2.529  | 3.163  | 3.734  | 4.438  |
|---------------------|-------|--------|--------|---------|--------|--------|--------|--------|
| fdi_percent_gdp     | 179.0 | 4.387  | 2.722  | -4.974  | 2.654  | 4.106  | 5.861  | 15.779 |
| gdp_growth          | 179.0 | 3.975  | 3.938  | -17.821 | 2.575  | 3.934  | 5.29   | 16.467 |
| gdp_per_capita_log  | 179.0 | 8.339  | 0.706  | 6.922   | 7.799  | 8.299  | 8.828  | 9.841  |
| inflation           | 179.0 | 5.317  | 5.058  | -1.55   | 2.676  | 4.481  | 6.967  | 51.461 |
| exports_percent_gdp | 179.0 | 33.998 | 14.653 | 13.522  | 23.95  | 30.592 | 42.312 | 78.299 |
| population_log      | 179.0 | 15.988 | 0.762  | 14.918  | 15.439 | 15.699 | 16.239 | 17.773 |
| unemployment        | 179.0 | 6.455  | 3.089  | 2.143   | 4.331  | 5.967  | 7.716  | 20.52  |

Nota: N = número de observaciones no faltantes.

### Panel Overview

Figura 1 — Visión general del panel

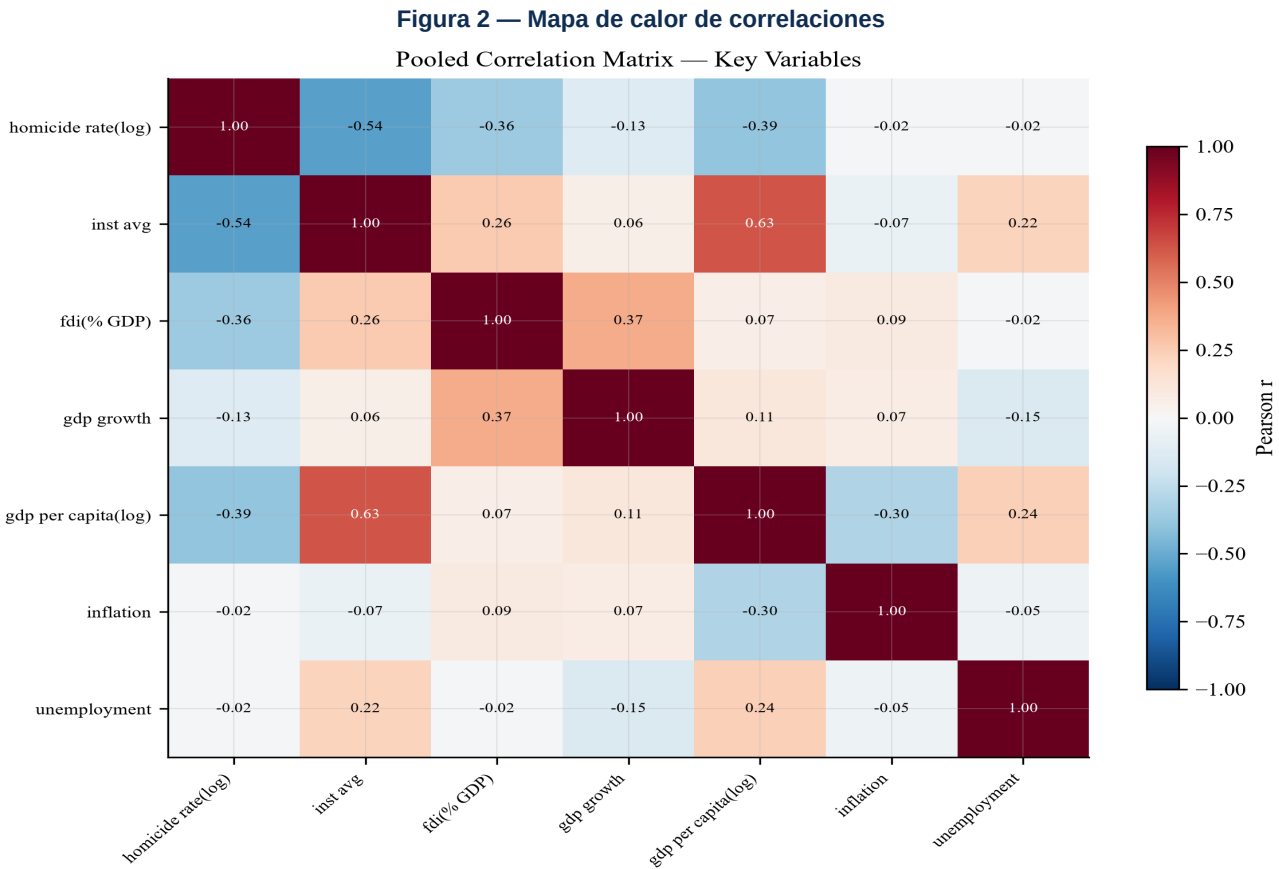
Panel Overview — Key Variables by Country (2000–2024)



Series temporales de las variables clave por país. Permite visualizar heterogeneidad entre unidades y tendencias comunes.

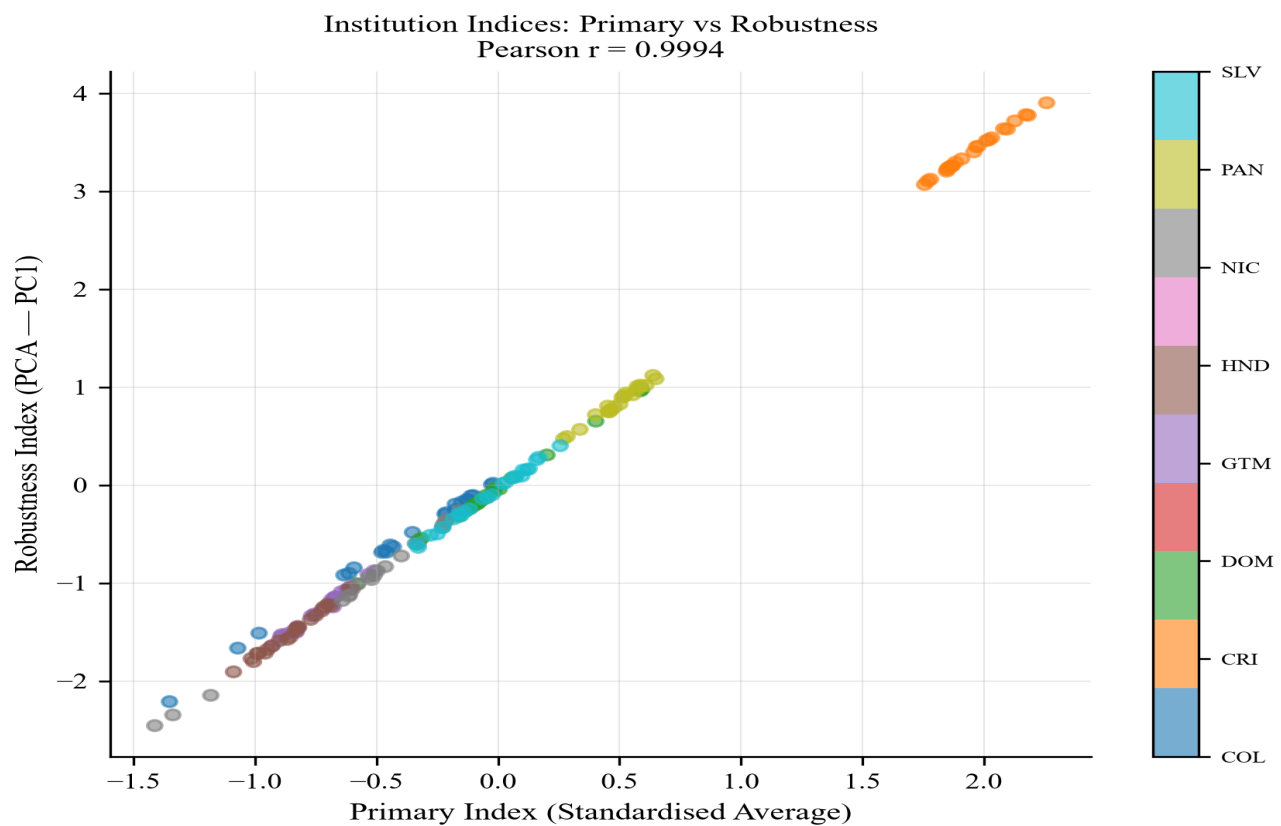
5. Correlaciones

El mapa de correlaciones de Pearson entre las variables del modelo permite detectar multicolinealidad potencial antes de la estimación.



Correlaciones de Pearson entre las principales variables del panel. Valores cercanos a  $\pm 1$  indican alta asociación lineal.

Figura 3 — Índices institucionales por país



*Evolución temporal de los índices institucionales (WGI) por unidad.*

## 6. Análisis de Componentes Principales (PCA)

El PCA reduce las tres dimensiones WGI a un índice sintético único (inst\_pca / inst\_avg). La construcción del índice sigue el procedimiento estándar: estandarización Z, extracción de PC1, corrección de signo para interpretación positiva.

|                            |        |                  |
|----------------------------|--------|------------------|
| Varianza explicada PC1     | 0.789  | > 0.60 deseable  |
| Alfa de Cronbach           | 0.863  | > 0.70 aceptable |
| KMO                        | 0.534  | > 0.50 mínimo    |
| Corr. inst_avg vs inst_pca | 0.9994 | > 0.95 muy alta  |

|                     |        |       |
|---------------------|--------|-------|
| rule_of_law         | 0.6300 | 36.5% |
| control_corruption  | 0.5865 | 34.0% |
| political_stability | 0.5091 | 29.5% |

### ■ Justificación del índice institucional

La alta correlación entre inst\_avg e inst\_pca ( $r = 0.9994$ ) indica que ambas medidas son prácticamente equivalentes. Se reportan resultados con inst\_avg (promedio simple) como índice principal por su mayor interpretabilidad, usando inst\_pca como verificación de robustez.

## 7. Modelo 1 — Violencia → Instituciones

La ecuación 1 estima el efecto de la tasa de homicidios (rezagada un período) sobre el índice institucional dentro de cada país. El uso del lag t-1 mitiga el sesgo de simultaneidad. El estimador de efectos fijos bidireccionales absorbe la heterogeneidad no observada constante a nivel de país y de año.

|                        |                                |        |        |  |        |        |
|------------------------|--------------------------------|--------|--------|--|--------|--------|
| homicide_rate_log_lag1 | -0.1404                        | 0.1358 | 0.3032 |  | 0.1282 | 0.2753 |
| Dep. variable          | inst_avg                       |        |        |  |        |        |
| N = 171                | R <sup>2</sup> within = 0.0738 |        |        |  |        |        |

### ■ Interpretación — EQ1

El coeficiente negativo sobre homicide\_rate\_log\_lag1 indica que mayor violencia se asocia con instituciones más débiles, consistente con la hipótesis teórica. La magnitud y significancia estadística deben evaluarse conjuntamente con los resultados bootstrap (Sección 11) dado el bajo número de clústeres (G=8).

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). \*\*\* p<0.01 \*\* p<0.05 \* p<0.10.

## 8. Modelo 2 — Instituciones → FDI

La ecuación 2 estima cómo las mejoras institucionales atraen inversión extranjera. Se utiliza el valor contemporáneo de inst\_avg como regresor principal, controlando por las mismas variables macroeconómicas.

|               |                                |        |        |   |        |        |
|---------------|--------------------------------|--------|--------|---|--------|--------|
| inst_avg      | 0.9000                         | 0.4648 | 0.0548 | * | 0.5690 | 0.1159 |
| Dep. variable | fdi_percent_gdp                |        |        |   |        |        |
| N = 179       | R <sup>2</sup> within = 0.2474 |        |        |   |        |        |

### ■ Interpretación — EQ2

Un coeficiente positivo sobre inst\_avg confirma el canal institucional: mejores instituciones reducen los riesgos de apropiación y contratos incompletos, atrayendo más IED. El p-valor marginal (~0.055) sugiere evidencia moderada.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). \*\*\* p<0.01 \*\* p<0.05 \* p<0.10.

## 9. Modelo 3 — FDI + Instituciones → Crecimiento

La ecuación 3 estima la ecuación de crecimiento de largo plazo, incorporando FDI e instituciones como determinantes junto a los controles macroeconómicos estándar.

| fdi_percent_gdp     | 0.1371                         | 0.2126 |     | 0.1371  | 0.5075 |
|---------------------|--------------------------------|--------|-----|---------|--------|
| inst_avg            | 0.5133                         | 0.7464 |     | 0.5133  | 0.6684 |
| gdp_per_capita_log  | 0.8887                         | 0.4857 |     | 0.8887  | 0.5015 |
| inflation           | -0.1126                        | 0.0002 | *** | -0.1126 | 0.0052 |
| exports_percent_gdp | 0.0635                         | 0.0987 | *   | 0.0635  | 0.2141 |
| population_log      | -0.7333                        | 0.9092 |     | -0.7333 | 0.8929 |
| unemployment        | -0.2101                        | 0.0642 | *   | -0.2101 | 0.0192 |
| Dep. variable       | gdp_growth                     |        |     |         |        |
| N = 179             | R <sup>2</sup> within = 0.0949 |        |     |         |        |

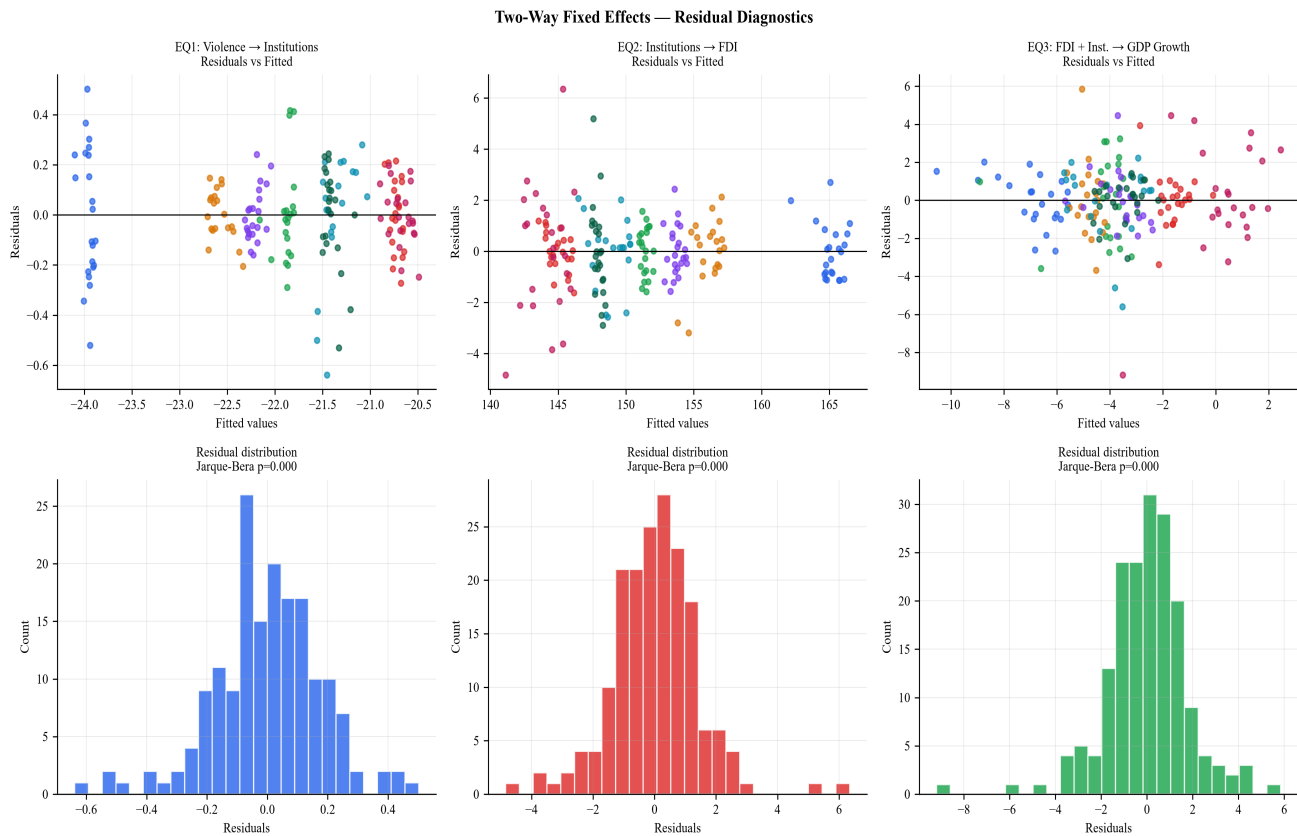
### ■ Interpretación — EQ3

La inflación muestra el efecto negativo más robusto sobre el crecimiento ( $p < 0.001$ ). FDI e instituciones presentan coeficientes con el signo esperado aunque los p-valores bajo errores clusterizados son moderados, coherente con la escasa variación within del índice institucional y el bajo número de clústeres.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). \*\*\*  $p < 0.01$  \*\*  $p < 0.05$  \*  $p < 0.10$ .

## Residuos de los modelos FE

Figura 4 — Diagnóstico visual de residuos (FE)



Gráficos de residuos para las tres ecuaciones. Permite inspeccionar heterocedasticidad y outliers.

## 10. Diagnósticos del Panel

Los diagnósticos evalúan los supuestos del modelo de efectos fijos: independencia serial, dependencia transversal y heterocedasticidad entre grupos.

### Ecuación 1

|                                      |                   |          |                                       |
|--------------------------------------|-------------------|----------|---------------------------------------|
| Test de Wooldridge (autocorrelación) | t = 4.760         | 4.42e-06 | Sí — autocorrelación serial detectada |
| Test de Pesaran (dep. transversal)   | CD = -66.722      | 0.0000   | Sí — dependencia cross-seccional      |
| Wald Modificado (heterocedasticidad) | W = 38.001 (df=7) | 3.03e-06 | Sí — heterocedasticidad de grupo      |

Nota:  $H_0$  varía por test. Ver descripción metodológica.

|                        |        |                |
|------------------------|--------|----------------|
| homicide_rate_log_lag1 | 1.3386 | ✓ Aceptable    |
| gdp_per_capita_log     | 1.3039 | ✓ Aceptable    |
| inflation              | 1.1972 | ✓ Aceptable    |
| exports_percent_gdp    | 1.6083 | ✓ Aceptable    |
| population_log         | 1.4258 | ✓ Aceptable    |
| unemployment           | 1.4780 | ✓ Aceptable    |
| Número de condición    | 2.3832 | < 30 aceptable |

### Ecuación 2

|                                      |                   |          |                                       |
|--------------------------------------|-------------------|----------|---------------------------------------|
| Test de Wooldridge (autocorrelación) | t = 2.066         | 0.0404   | Sí — autocorrelación serial detectada |
| Test de Pesaran (dep. transversal)   | CD = -64.770      | 0.0000   | Sí — dependencia cross-seccional      |
| Wald Modificado (heterocedasticidad) | W = 67.638 (df=7) | 4.43e-12 | Sí — heterocedasticidad de grupo      |

Nota:  $H_0$  varía por test. Ver descripción metodológica.

|                     |        |                 |
|---------------------|--------|-----------------|
| inst_avg            | 1.1945 | ✓ Acceptable    |
| gdp_per_capita_log  | 1.3350 | ✓ Acceptable    |
| inflation           | 1.1780 | ✓ Acceptable    |
| exports_percent_gdp | 1.6773 | ✓ Acceptable    |
| population_log      | 1.4152 | ✓ Acceptable    |
| unemployment        | 1.2186 | ✓ Acceptable    |
| Número de condición | 2.2647 | < 30 acceptable |

### Ecuación 3

|                                      |                   |          |                                       |
|--------------------------------------|-------------------|----------|---------------------------------------|
| Test de Wooldridge (autocorrelación) | t = 2.328         | 0.0211   | Sí — autocorrelación serial detectada |
| Test de Pesaran (dep. transversal)   | CD = -72.112      | 0.0000   | Sí — dependencia cross-seccional      |
| Wald Modificado (heterocedasticidad) | W = 40.945 (df=7) | 8.30e-07 | Sí — heterocedasticidad de grupo      |

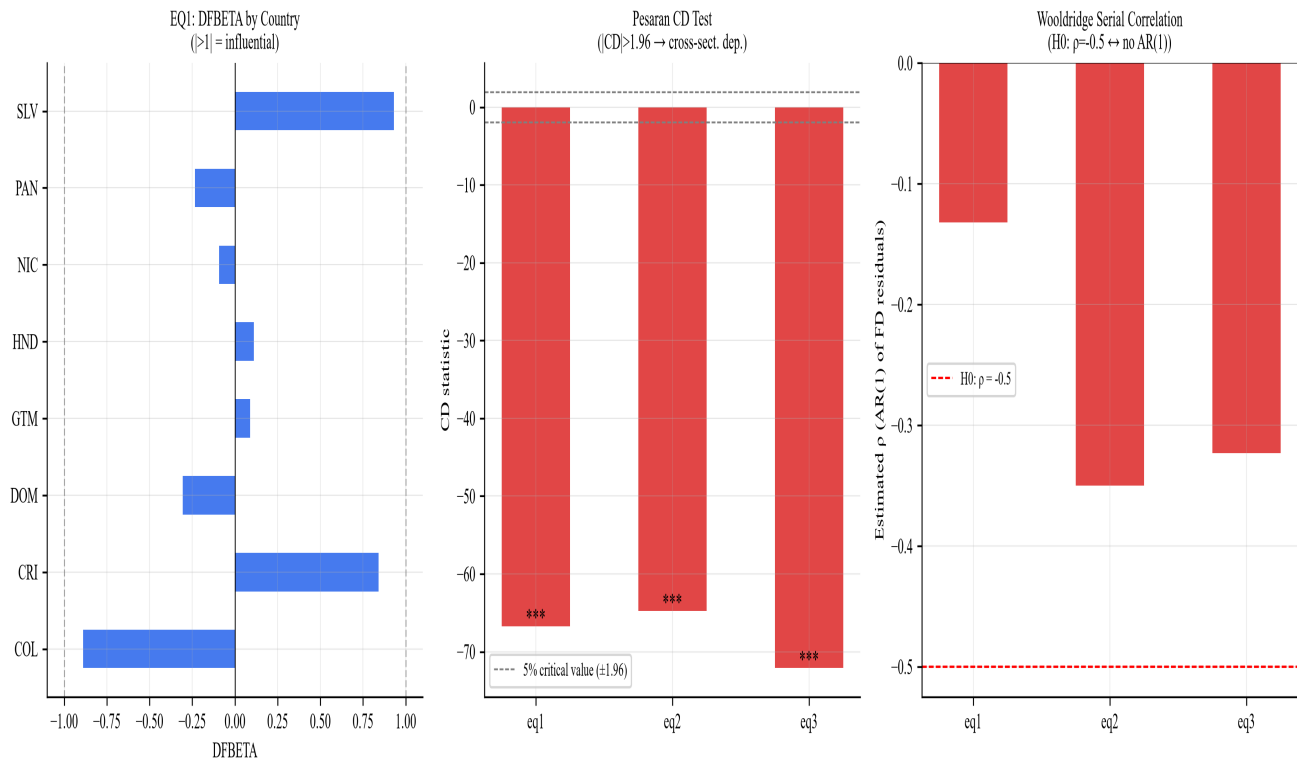
Nota: H<sub>0</sub> varía por test. Ver descripción metodológica.

|                     |        |                 |
|---------------------|--------|-----------------|
| fdi_percent_gdp     | 1.3216 | ✓ Acceptable    |
| inst_avg            | 1.2113 | ✓ Acceptable    |
| gdp_per_capita_log  | 1.3788 | ✓ Acceptable    |
| inflation           | 1.1965 | ✓ Acceptable    |
| exports_percent_gdp | 1.8892 | ✓ Acceptable    |
| population_log      | 1.4859 | ✓ Acceptable    |
| unemployment        | 1.2639 | ✓ Acceptable    |
| Número de condición | 2.5609 | < 30 acceptable |

#### ■ Implicaciones para la inferencia

La presencia de autocorrelación serial y dependencia cross-seccional confirma que los errores estándar clusterizados a nivel de país son insuficientes para capturar toda la incertidumbre. Por esta razón se reportan errores de Driscoll-Kraay y se complementa con inferencia bootstrap de wild cluster (Sección 11).

Figura 5 — Diagnósticos panel  
Panel Diagnostic Summary



Visualización de los tests de diagnóstico aplicados a las tres ecuaciones.

## 11. Inferencia Bootstrap (Wild Cluster)

Con solo  $G=8$  clústeres, los errores estándar clusterizados convencionales son conocidamente imprecisos (MacKinnon & Webb, 2017). Se aplica el wild cluster bootstrap con pesos de Webb (2023) usando  $B=999$  réplicas. El p-valor bootstrap es exacto por construcción bajo  $H_0$ .

| EQ1: Violencia → Inst. | -0.1404 | 0.1358 | -1.034 | 0.4294 | [-0.2041, 0.2068] | 999 |
|------------------------|---------|--------|--------|--------|-------------------|-----|
| EQ2: Inst. → FDI       | 0.9000  | 0.4648 | 1.936  | 0.0621 | [-0.8424, 0.8392] | 999 |
| EQ3: FDI → Crecimiento | 0.1371  | 0.1095 | 1.252  | 0.4094 | [-0.2404, 0.2412] | 999 |

### Errores CR2 (Bell-McCaffrey) — EQ1: Violencia → Inst.

| homicide_rate_log_lag1 | 0.1420 | 0.3922 |  |
|------------------------|--------|--------|--|
| gdp_per_capita_log     | 0.2082 | 0.4818 |  |
| inflation              | 0.0054 | 0.6730 |  |
| exports_percent_gdp    | 0.0100 | 0.3770 |  |
| population_log         | 2.2180 | 0.6201 |  |
| unemployment           | 0.0116 | 0.4546 |  |

### Errores CR2 (Bell-McCaffrey) — EQ2: Inst. → FDI

| inst_avg            | 0.5155 | 0.2023 |    |
|---------------------|--------|--------|----|
| gdp_per_capita_log  | 1.2402 | 0.2276 |    |
| inflation           | 0.0185 | 0.0722 | *  |
| exports_percent_gdp | 0.0138 | 0.0128 | ** |
| population_log      | 4.7400 | 0.1092 |    |
| unemployment        | 0.1438 | 0.2712 |    |

### Errores CR2 (Bell-McCaffrey) — EQ3: FDI → Crecimiento

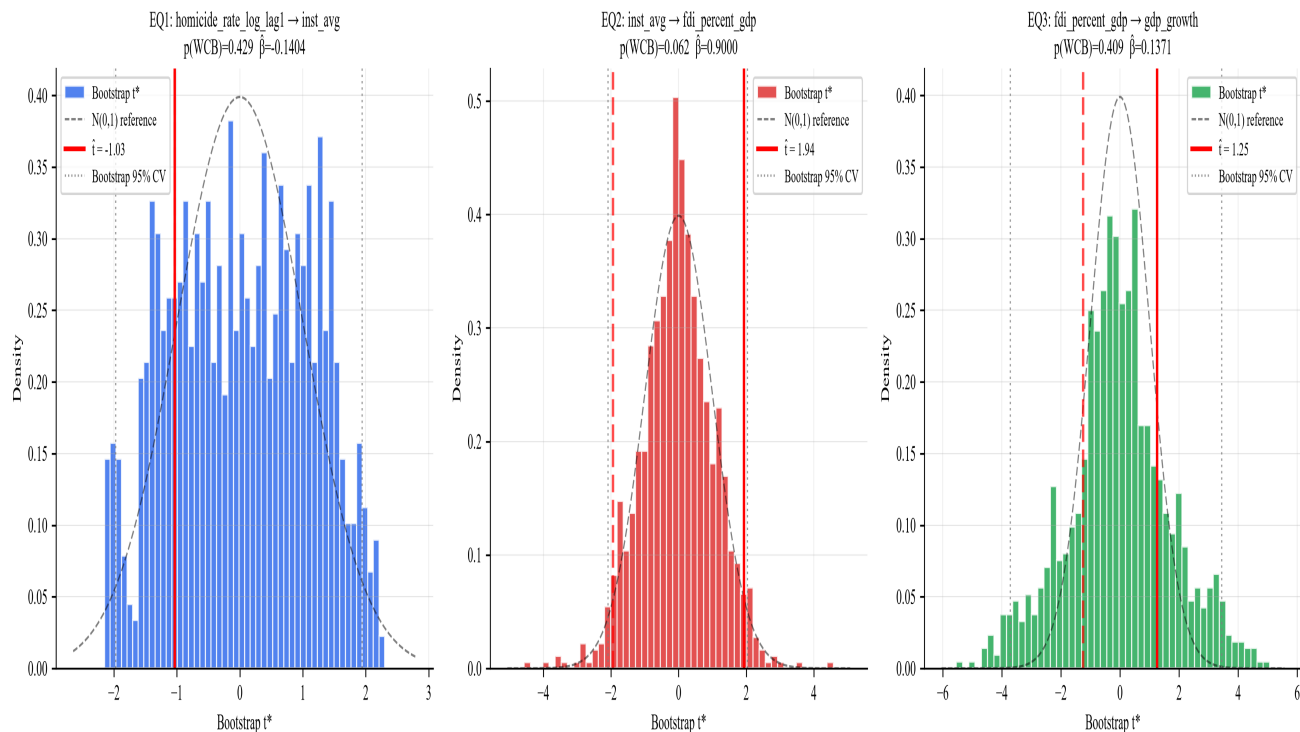
|                     |        |        |    |
|---------------------|--------|--------|----|
| fdi_percent_gdp     | 0.1133 | 0.3618 |    |
| inst_avg            | 1.8324 | 0.8107 |    |
| gdp_per_capita_log  | 1.4638 | 0.6011 |    |
| inflation           | 0.0359 | 0.0367 | ** |
| exports_percent_gdp | 0.0477 | 0.3060 |    |
| population_log      | 8.5839 | 0.9424 |    |
| unemployment        | 0.1234 | 0.1791 |    |

### ■ Wild Cluster Bootstrap — Nota metodológica

Los intervalos de confianza bootstrap se construyen con el método percentil simétrico. Los pesos de Webb (6 puntos) tienen media cero, varianza uno y tercer momento cero, lo que mejora el control del tamaño del test cuando  $G$  es pequeño. Un  $p\text{-boot} > 0.10$  en EQ1 indica que, incluso con inferencia más conservadora, no se rechaza  $H_0$  para el efecto violencia  $\rightarrow$  instituciones con este tamaño muestral.

**Figura 6 — Distribuciones bootstrap**

Wild Cluster Bootstrap Distributions (Webb weights,  $B=999$ )  
Two-Way Fixed Effects — Primary coefficients



Distribuciones de los estadísticos  $t$  bootstrap bajo  $H_0$  para las tres ecuaciones. La línea vertical indica el  $t$  observado.

## 12. Pruebas de Robustez

Los análisis de robustez verifican la estabilidad de los coeficientes ante: (i) exclusión de cada país (LOCO — Leave-One-Country-Out), (ii) especificaciones alternativas del índice institucional, (iii) variaciones en las variables de control.

### LOCO — EQ1 (Violencia → Instituciones)

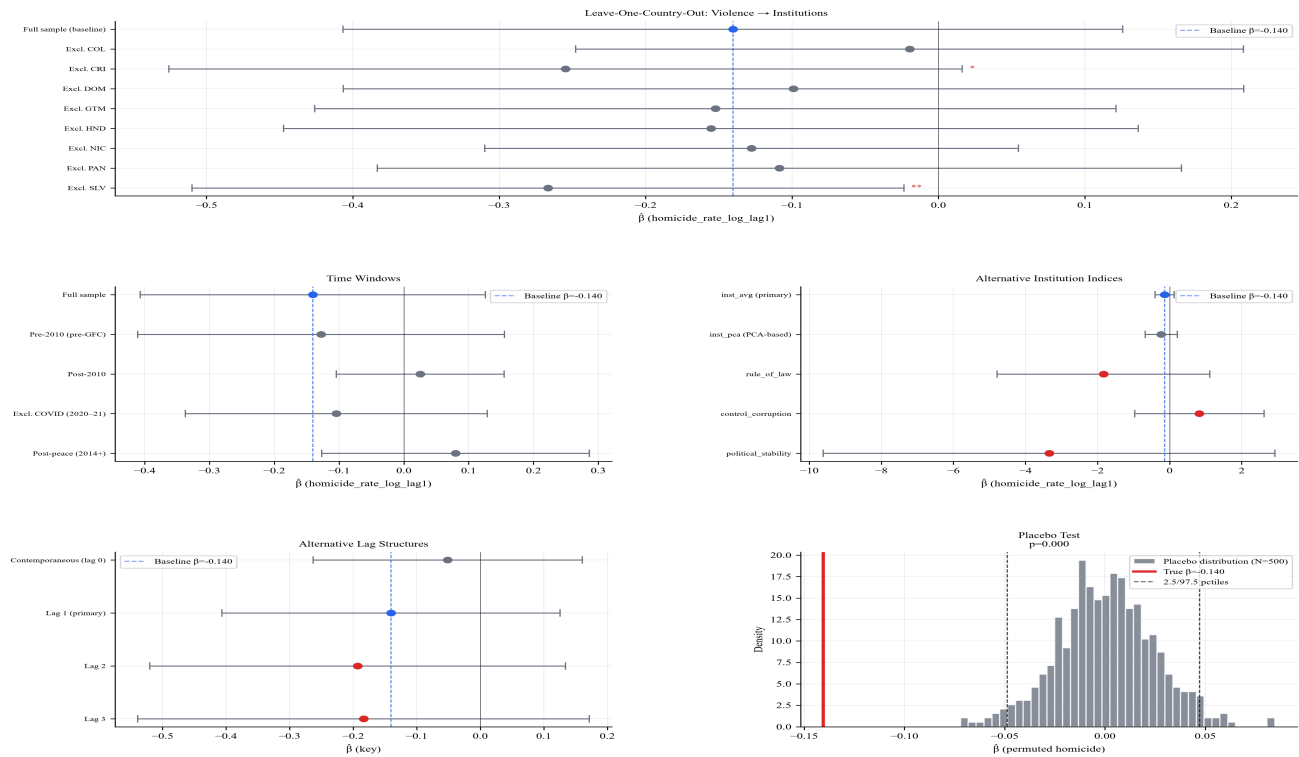
| Full sample (baseline) | -0.1404 | 0.1358 | 0.3032 | 0.0738  | 171 |
|------------------------|---------|--------|--------|---------|-----|
| Excl. COL              | -0.0197 | 0.1164 | 0.8660 | 0.1750  | 149 |
| Excl. CRI              | -0.2547 | 0.1383 | 0.0681 | 0.0454  | 149 |
| Excl. DOM              | -0.0990 | 0.1569 | 0.5295 | 0.0044  | 150 |
| Excl. GTM              | -0.1523 | 0.1396 | 0.2775 | 0.1410  | 151 |
| Excl. HND              | -0.1554 | 0.1490 | 0.2991 | 0.0984  | 149 |
| Excl. NIC              | -0.1277 | 0.0930 | 0.1723 | 0.1040  | 152 |
| Excl. PAN              | -0.1087 | 0.1401 | 0.4395 | 0.1925  | 149 |
| Excl. SLV              | -0.2668 | 0.1241 | 0.0336 | -8.5015 | 148 |

#### ■ LOCO — Estabilidad del coeficiente

Si el coeficiente permanece negativo y de magnitud similar al excluir cada país, se concluye que los resultados no están impulsados por ninguna observación influyente. Una variación sustancial al excluir un país señala dependencia de ese clúster.

Figura 7 — Robustez

Robustness Analysis — Coefficient Stability  
Two-Way Fixed Effects | Dep. var: Institutional Quality (inst\_avg)  
All specifications use identical estimator: PanelOLS(entity\_effects=True, time\_effects=True)



Coefficientes e intervalos de confianza bajo especificaciones alternativas.

### 13. Triangulación Machine Learning

El análisis de ML (Random Forest y Gradient Boosting con LOCO-CV) sirve como triangulación no paramétrica del ranking de importancia de variables. No reemplaza la inferencia causal del modelo econométrico: su objetivo es confirmar que las variables clave aparecen consistentemente como predictores relevantes.

#### Tarea ML — T1: Violence → Institutions

|                                |          |                                                                |
|--------------------------------|----------|----------------------------------------------------------------|
| CV R <sup>2</sup> (media)      | -33.1112 | -49.0544                                                       |
| CV R <sup>2</sup> (desv. est.) | 49.1064  | 74.8629                                                        |
| Wilcoxon RF vs GB (p)          | 0.8438   | G=8 pairs → very low power. Treat p-value as descriptive only. |

|                        |        |        |        |      |
|------------------------|--------|--------|--------|------|
| gdp_per_capita_log     | 0.1676 | 0.2029 | 0.1752 | 2.52 |
| homicide_rate_log_lag1 | 0.1077 | 0.1314 | 0.0721 | 4.34 |
| homicide_rate_log_lag2 | 0.1020 | 0.1264 | 0.0703 | 3.17 |
| population_log         | 0.2440 | 0.3159 | 0.3473 | 1.61 |
| unemployment           | 0.0650 | 0.0789 | 0.0576 | 5.34 |

#### ■ Interpretación ML — EQ1

Variable clave rango SHAP: #3. Rango permutación: #3. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

#### Tarea ML — T2: Institutions → FDI

|                                |         |                                                                |
|--------------------------------|---------|----------------------------------------------------------------|
| CV R <sup>2</sup> (media)      | -0.4564 | -0.7026                                                        |
| CV R <sup>2</sup> (desv. est.) | 0.7686  | 1.0402                                                         |
| Wilcoxon RF vs GB (p)          | 0.4609  | G=8 pairs → very low power. Treat p-value as descriptive only. |

| exports_percent_gdp    | 0.7221 | 0.3082 | 0.3099 | 1.10 |
|------------------------|--------|--------|--------|------|
| gdp_per_capita_log     | n/a    | 0.0905 | n/a    | 4.67 |
| homicide_rate_log_lag1 | 0.3994 | 0.1420 | 0.0962 | 3.14 |
| inst_avg               | 0.2509 | 0.1106 | 0.0741 | 4.23 |
| tourist_arrivals_log   | 0.1661 | n/a    | 0.0624 | 5.14 |
| unemployment           | 0.2487 | 0.1618 | 0.1178 | 4.20 |

#### ■ Interpretación ML — EQ2

Variable clave rango SHAP: #3. Rango permutación: #4. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

### Tarea ML — T3: FDI + Institutions → Growth

|                                |        |                                                                |
|--------------------------------|--------|----------------------------------------------------------------|
| CV R <sup>2</sup> (media)      | 0.0989 | 0.2418                                                         |
| CV R <sup>2</sup> (desv. est.) | 0.1458 | 0.4056                                                         |
| Wilcoxon RF vs GB (p)          | 0.3828 | G=8 pairs → very low power. Treat p-value as descriptive only. |

| exports_percent_gdp | 0.3021 | 0.1438 | 0.0743 | 3.02 |
|---------------------|--------|--------|--------|------|
| fdi_percent_gdp     | 0.3284 | 0.1833 | 0.1102 | 1.98 |
| gdp_per_capita_log  | 0.2009 | n/a    | n/a    | 5.20 |
| inflation           | 0.2243 | 0.1300 | 0.0677 | 4.30 |
| unemployment        | n/a    | 0.1173 | 0.0540 | 5.58 |
| year_c              | 0.3029 | 0.1988 | 0.1760 | 4.17 |

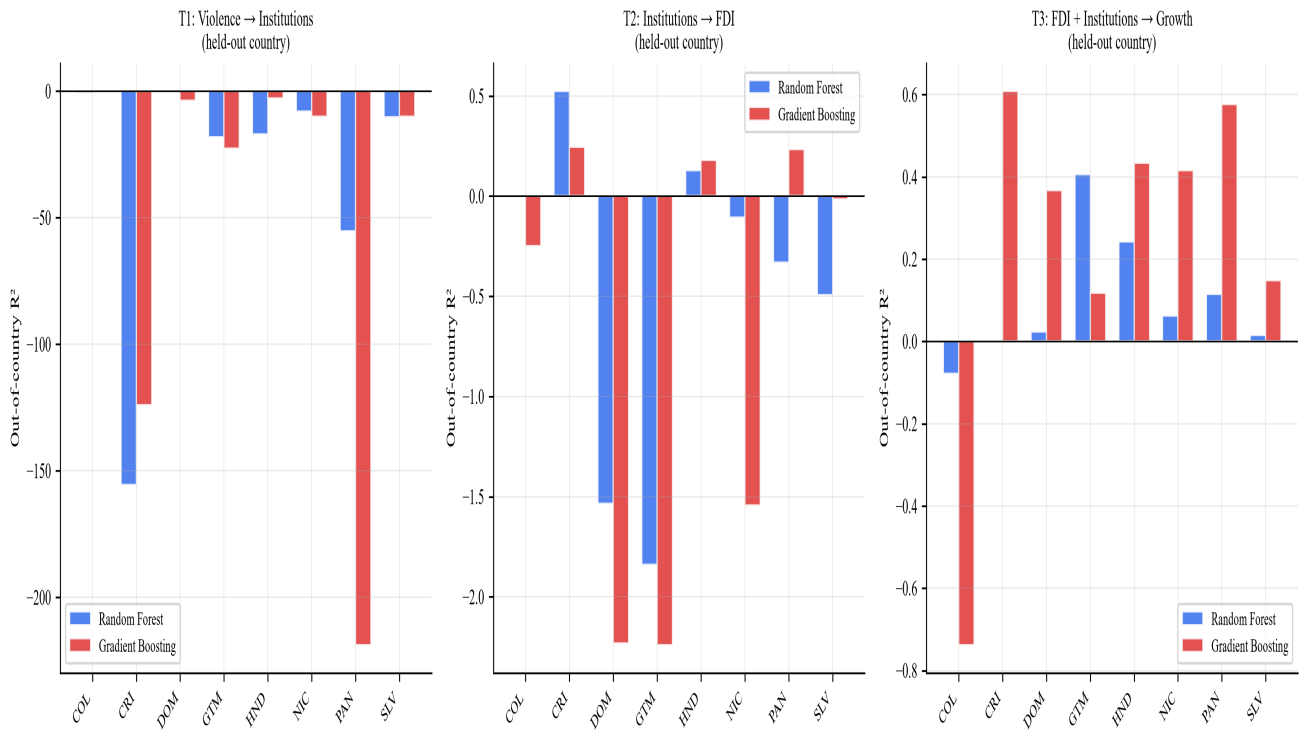
#### ■ Interpretación ML — EQ3

Variable clave rango SHAP: #1. Rango permutación: #2. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Figura 8 — LOCO-CV  $R^2$  por país

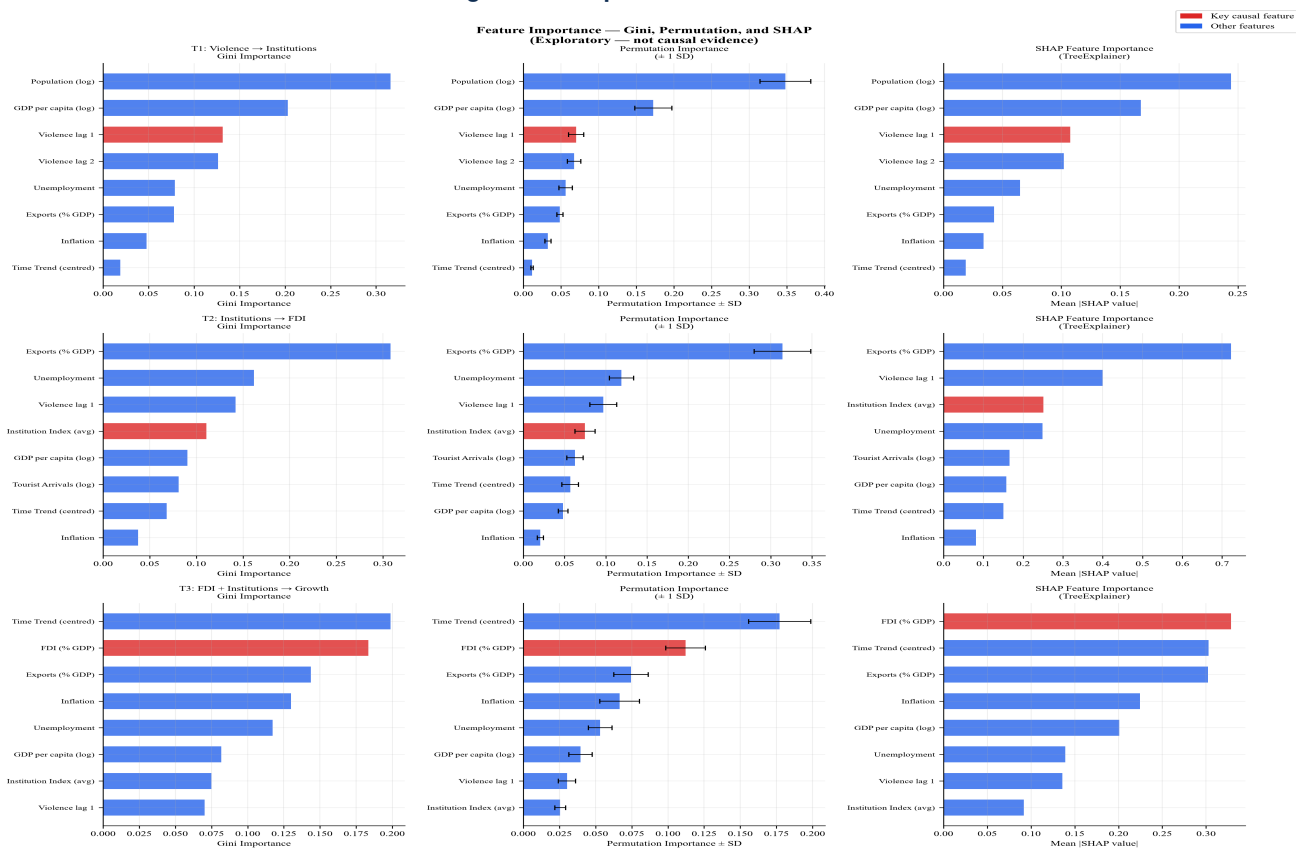
Leave-One-Country-Out Cross-Validation (LOCO-CV)

Random Forest & Gradient Boosting | Note: G=8 folds → wide uncertainty



$R^2$  de validación cruzada leave-one-country-out para RF y GB en las tres tareas.

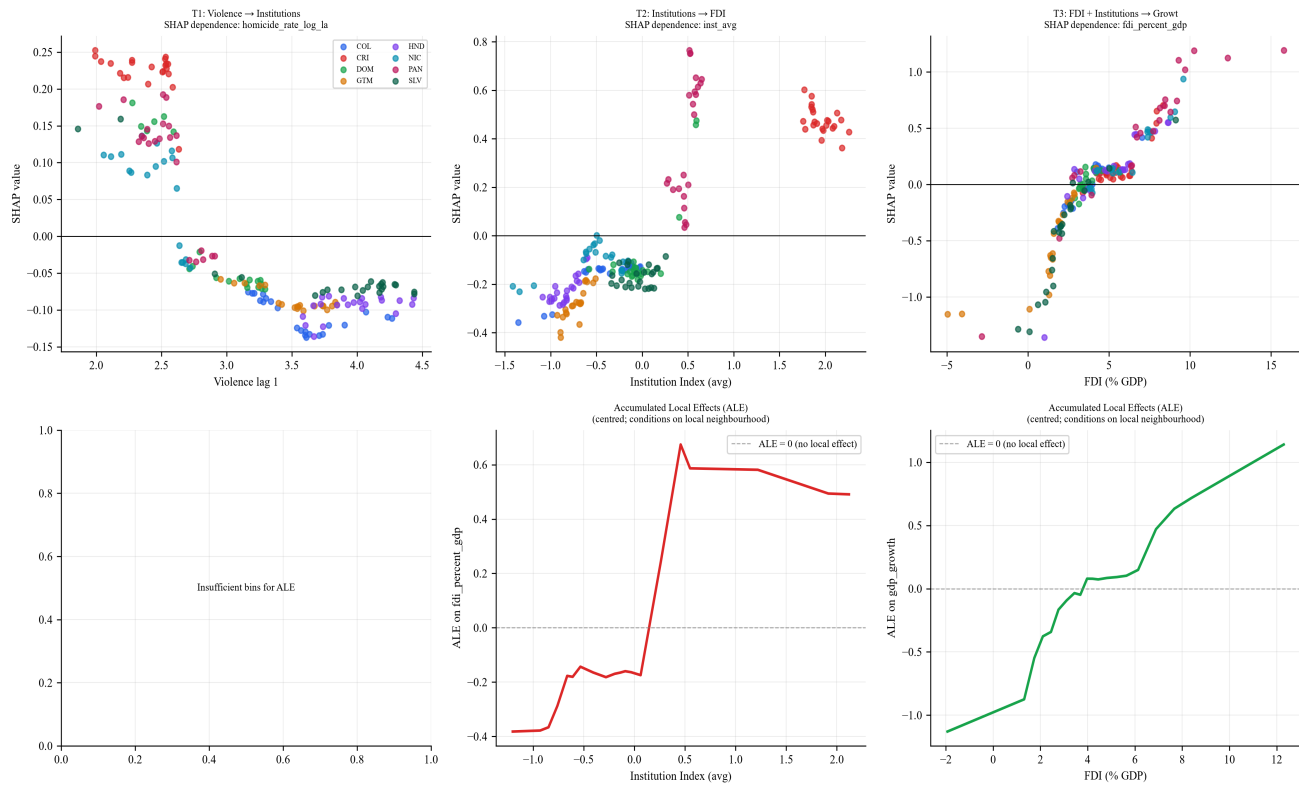
Figura 9 — Importancia de variables



Importancia Gini, permutación y SHAP de las variables en las tres tareas ML.

Figura 10 — SHAP / ALE plots

SHAP Dependence Plots & ALE Plots — Key Causal Variables  
(Exploratory — shows non-linearity, not causal effect)  
ALE used instead of PDP due to correlated predictors (Apley & Zhu 2020)



Efectos marginales acumulados (ALE) y valores SHAP para los predictores clave.

## 14. Conclusiones

A continuación se sintetizan las principales conclusiones derivadas estrictamente de los resultados calculados. No se realizan inferencias extrapoladas.

|   |                                                                              |                                                                                      |
|---|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1 | Violencia → Instituciones (EQ1): coeficiente negativo, no significativo      | $\beta = -0.1404$ , p-CL = 0.3032, p-boot = 0.4294                                   |
| 2 | Instituciones → FDI (EQ2): coeficiente positivo, marginalmente significativo | $\beta = 0.9000$ , p-CL = 0.0548, p-boot = 0.0621                                    |
| 3 | FDI → Crecimiento (EQ3): coeficiente positivo, no significativo              | $\beta = 0.1371$ , p-CL = 0.2126, p-boot = 0.4094                                    |
| 4 | Inflación → Crecimiento: el único efecto estadísticamente robusto en EQ3     | $\beta = -0.1126$ , p-CL = 0.0002                                                    |
| 5 | Bajo número de clústeres (G=8) limita el poder estadístico                   | Bootstrap confirma amplios intervalos de confianza para todos los coeficientes clave |

### ■ Limitaciones metodológicas

Con G=8 clústeres, el poder de los tests convencionales es reducido. La no significancia estadística no implica ausencia de efecto económico — los intervalos de confianza bootstrap son amplios y compatibles tanto con efectos nulos como con efectos moderados. Se recomienda ampliar la muestra geográfica para obtener inferencia más precisa.





NIC: missing years [2001, 2020, 2022, 2023, 2024]

PAN: missing years [2001, 2024]

PAN: missing years [2001, 2024]

SLV: missing years [2001]

SLV: missing years [2001]

■■ Column validation ■■

■■ Column validation ■■

✓ All 21 required columns present.

✓ All 21 required columns present.

■■ Missingness ■■

■■ Missingness ■■

■ Missing values:

■ Missing values:

trade\_percent\_gdp 179 (100.0%)

trade\_percent\_gdp 179 (100.0%)

homicide\_rate\_log\_lag3 24 (13.4%)

homicide\_rate\_log\_lag3 24 (13.4%)

homicide\_rate\_log\_lag3 24 (13.4%)

homicide\_rate\_log\_lag3 24 (13.4%)

tourist\_arrivals\_log 23 (12.8%)

tourist\_arrivals\_log 23 (12.8%)

homicide\_rate\_log\_lag2 16 (8.9%)

homicide\_rate\_log\_lag2 16 (8.9%)

homicide\_rate\_log\_lag2 16 (8.9%)

homicide\_rate\_log\_lag2 16 (8.9%)

homicide\_rate\_log\_lag1 8 (4.5%)

homicide\_rate\_log\_lag1 8 (4.5%)

homicide\_rate\_log\_lag1 8 (4.5%)

homicide\_rate\_log\_lag1 8 (4.5%)

02 — CENTRED TIME TREND

02 — CENTRED TIME TREND

✓ year\_c = year – 2011.8 (range: -12 to 12)

✓ year\_c = year – 2011.8 (range: -12 to 12)

Rationale: centring eliminates the large-value component of year that

Rationale: centring eliminates the large-value component of year that

inflates the condition number of the design matrix (collinearity).

inflates the condition number of the design matrix (collinearity).

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

All three dimensions are first z-scored (global mean=0, SD=1),

All three dimensions are first z-scored (global mean=0, SD=1),

then averaged. This ensures equal weighting regardless of raw scale.

then averaged. This ensures equal weighting regardless of raw scale.

✓ *inst\_avg*: mean=-0.0000, SD=0.8882, range=[-1.414, 2.259]  
 ✓ *inst\_avg*: mean=-0.0000, SD=0.8882, range=[-1.414, 2.259]  
 Higher *inst\_avg* → better governance (all three WGI dimensions).  
 Higher *inst\_avg* → better governance (all three WGI dimensions).  
 Cronbach's  $\alpha$  (internal consistency of index): ✓  $\alpha = 0.863$  (good internal consistency)  
 Cronbach's  $\alpha$  (internal consistency of index): ✓  $\alpha = 0.863$  (good internal consistency)

04 — INSTITUTION INDEX: ROBUSTNESS (PCA)  
04 — INSTITUTION INDEX: ROBUSTNESS (PCA)

PCA operates on the same three z-scored variables.  
 PCA operates on the same three z-scored variables.  
 PC1 is retained. Sign is aligned so higher = better governance.  
 PC1 is retained. Sign is aligned so higher = better governance.  
 ■ KMO = 0.534 &lt; 0.6 — marginal adequacy  
 ■ KMO = 0.534 &lt; 0.6 — marginal adequacy  
 rule\_of\_law KMO = 0.520  
 rule\_of\_law KMO = 0.520  
 control\_corruption KMO = 0.531  
 control\_corruption KMO = 0.531  
 political\_stability KMO = 0.570  
 political\_stability KMO = 0.570  
 Bartlett sphericity: ✓  $\chi^2(3)=406.8, p=0.0000$  — reject H0, PCA appropriate  
 Bartlett sphericity: ✓  $\chi^2(3)=406.8, p=0.0000$  — reject H0, PCA appropriate  
 ✓ PC1 explains 78.9% of variance in the three WGI dimensions.  
 ✓ PC1 explains 78.9% of variance in the three WGI dimensions.  
 PC2: 18.5%, PC3: 2.5%  
 PC2: 18.5%, PC3: 2.5%

PCA loadings (PC1):

rule\_of\_law 0.6300 ████████████████████████████████

control\_corruption 0.5865 ████████████████████████████████

political\_stability 0.5091 ████████████████████████████████

✓ Correlation(inst\_avg, inst\_pca) = 0.9994 — high convergence

✓ Correlation(inst\_avg, inst\_pca) = 0.9994 — high convergence

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

Reference: Wooldridge (2010), *Econometric Analysis of Cross Section  
Reference: Wooldridge (2010), Econometric Analysis of Cross Section  
and Panel Data*, 2nd ed., ch. 10.  
and Panel Data, 2nd ed., ch. 10.  
'Within' variance is the variation exploited by Fixed Effects.  
'Within' variance is the variation exploited by Fixed Effects.  
Variable Total SD Between SD Within SD Within %

Variable Total SD Between SD Within SD Within %

homicide\_rate\_log 0.6981 0.6257 0.3748 53.7% ✓  
 homicide\_rate\_log 0.6981 0.6257 0.3748 53.7% ✓  
 inst\_avg 0.8882 0.9108 0.2219 25.0% ✓  
 inst\_avg 0.8882 0.9108 0.2219 25.0% ✓  
 inst\_pca 1.5432 1.5847 0.3786 24.5% ✓  
 inst\_pca 1.5432 1.5847 0.3786 24.5% ✓  
 fdi\_percent\_gdp 2.7215 1.9499 2.0259 74.4% ✓  
 fdi\_percent\_gdp 2.7215 1.9499 2.0259 74.4% ✓  
 gdp\_growth 3.9384 1.1250 3.7888 96.2% ✓  
 gdp\_growth 3.9384 1.1250 3.7888 96.2% ✓  
 gdp\_per\_capita\_log 0.7064 0.6468 0.3705 52.4% ✓  
 gdp\_per\_capita\_log 0.7064 0.6468 0.3705 52.4% ✓

■ Variables with low within % are primarily identified from cross-section.

■ Variables with low within % are primarily identified from cross-section.

Fixed Effects may not efficiently estimate their coefficients.

Fixed Effects may not efficiently estimate their coefficients.

06 — SUMMARY STATISTICS

06 — SUMMARY STATISTICS

count mean std min 10% 50% 90% max

count mean std min 10% 50% 90% max

Variable

Variable

homicide\_rate 179.0 28.388 20.806 1.900 8.724 22.643 60.522 107.638  
 homicide\_rate 179.0 28.388 20.806 1.900 8.724 22.643 60.522 107.638  
 homicide\_rate\_log 179.0 3.139 0.698 1.860 2.275 3.163 4.119 4.438  
 homicide\_rate\_log 179.0 3.139 0.698 1.860 2.275 3.163 4.119 4.438  
 rule\_of\_law 179.0 48.012 8.851 31.590 39.066 46.323 65.832 68.454  
 rule\_of\_law 179.0 48.012 8.851 31.590 39.066 46.323 65.832 68.454  
 control\_corruption 179.0 37.927 10.225 22.806 27.218 36.541 57.239 68.592  
 control\_corruption 179.0 37.927 10.225 22.806 27.218 36.541 57.239 68.592  
 political\_stability 179.0 61.261 11.386 25.320 49.719 62.311 76.807 86.123  
 political\_stability 179.0 61.261 11.386 25.320 49.719 62.311 76.807 86.123  
 inst\_avg 179.0 -0.000 0.888 -1.414 -0.872 -0.161 1.852 2.259  
 inst\_avg 179.0 -0.000 0.888 -1.414 -0.872 -0.161 1.852 2.259  
 inst\_pca 179.0 0.000 1.543 -2.456 -1.530 -0.272 3.223 3.904  
 inst\_pca 179.0 0.000 1.543 -2.456 -1.530 -0.272 3.223 3.904  
 fdi\_percent\_gdp 179.0 4.387 2.722 -4.974 1.517 4.106 7.961 15.779  
 fdi\_percent\_gdp 179.0 4.387 2.722 -4.974 1.517 4.106 7.961 15.779  
 gdp\_growth 179.0 3.975 3.938 -17.821 0.981 3.934 8.184 16.467  
 gdp\_growth 179.0 3.975 3.938 -17.821 0.981 3.934 8.184 16.467  
 gdp\_per\_capita\_log 179.0 8.339 0.706 6.922 7.451 8.299 9.403 9.841  
 gdp\_per\_capita\_log 179.0 8.339 0.706 6.922 7.451 8.299 9.403 9.841  
 inflation 179.0 5.317 5.058 -1.550 0.980 4.481 9.252 51.461  
 inflation 179.0 5.317 5.058 -1.550 0.980 4.481 9.252 51.461  
 unemployment 179.0 6.455 3.089 2.143 3.044 5.967 10.594 20.520  
 unemployment 179.0 6.455 3.089 2.143 3.044 5.967 10.594 20.520  
 exports\_percent\_gdp 179.0 33.998 14.653 13.522 17.484 30.592 53.997 78.299

```

exports_percent_gdp 179.0 33.998 14.653 13.522 17.484 30.592 53.997 78.299
population_log 179.0 15.988 0.762 14.918 15.225 15.699 17.559 17.773
population_log 179.0 15.988 0.762 14.918 15.225 15.699 17.559 17.773
tourist_arrivals_log 156.0 14.393 0.587 13.269 13.595 14.401 15.206 15.837
tourist_arrivals_log 156.0 14.393 0.587 13.269 13.595 14.401 15.206 15.837

```

✓ LaTeX summary table saved → tables/summary\_statistics.tex

✓ LaTeX summary table saved → tables/summary\_statistics.tex

07 — FIGURES

07 — FIGURES

✓ Figure saved → figures/01\_panel\_overview.png

✓ Figure saved → figures/01\_panel\_overview.png

✓ Figure saved → figures/02\_institution\_indices.png

✓ Figure saved → figures/02\_institution\_indices.png

✓ Figure saved → figures/03\_correlation\_heatmap.png

✓ Figure saved → figures/03\_correlation\_heatmap.png

08 — EXPORT

08 — EXPORT

✓ Enriched panel saved → panel\_enriched.csv (179 rows)

✓ Enriched panel saved → panel\_enriched.csv (179 rows)

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\01\_metadata.json

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\01\_metadata.json

Module 01 complete. Enriched dataset and metadata exported.

Module 01 complete. Enriched dataset and metadata exported.

✓ Module 01 completed in 21.9s

02-A — LOAD DATA

02-A — LOAD DATA

✓ Data loaded: 179 obs, 8 countries

✓ Data loaded: 179 obs, 8 countries

02-B — BENCHMARK: POOLED OLS

02-B — BENCHMARK: POOLED OLS

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

Reference: Angrist & Pischke (2009) — benchmark before FE.

Reference: Angrist & Pischke (2009) — benchmark before FE.

EQ1: Violence→Institutions (N=171, R<sup>2</sup>=0.7863)

EQ1: Violence→Institutions (N=171, R<sup>2</sup>=0.7863)

```

homicide_rate_log_lag1 -0.1369 p=0.145
homicide_rate_log_lag1 -0.1369 p=0.145
gdp_per_capita_log 0.7859 p=0.000***
gdp_per_capita_log 0.7859 p=0.000***
inflation 0.0174 p=0.283
inflation 0.0174 p=0.283
exports_percent_gdp -0.0164 p=0.009***
exports_percent_gdp -0.0164 p=0.009***
population_log -0.7506 p=0.000***
population_log -0.7506 p=0.000***
unemployment 0.0863 p=0.000***
unemployment 0.0863 p=0.000***
year_c -0.0333 p=0.002***
year_c -0.0333 p=0.002***
EQ2: Institutions→FDI (N=179, R²=0.4318)
EQ2: Institutions→FDI (N=179, R²=0.4318)
inst_avg 1.3126 p=0.000***
inst_avg 1.3126 p=0.000***
gdp_per_capita_log -0.8355 p=0.328
gdp_per_capita_log -0.8355 p=0.328
inflation 0.0078 p=0.789
inflation 0.0078 p=0.789
exports_percent_gdp 0.1446 p=0.000***
exports_percent_gdp 0.1446 p=0.000***
population_log 1.2946 p=0.014**
population_log 1.2946 p=0.014**
unemployment 0.0041 p=0.975
unemployment 0.0041 p=0.975
year_c 0.0369 p=0.554
year_c 0.0369 p=0.554
EQ3: FDI+Inst→GDP Growth (N=179, R²=0.2284)
EQ3: FDI+Inst→GDP Growth (N=179, R²=0.2284)
fdi_percent_gdp 0.4808 p=0.000***
fdi_percent_gdp 0.4808 p=0.000***
inst_avg -0.3664 p=0.342
inst_avg -0.3664 p=0.342
gdp_per_capita_log 1.6958 p=0.000***
gdp_per_capita_log 1.6958 p=0.000***
inflation 0.0314 p=0.610
inflation 0.0314 p=0.610
exports_percent_gdp 0.0412 p=0.018**
exports_percent_gdp 0.0412 p=0.018**
population_log 1.0786 p=0.021**
population_log 1.0786 p=0.021**
unemployment -0.2831 p=0.000***
unemployment -0.2831 p=0.000***
year_c -0.1014 p=0.005***
year_c -0.1014 p=0.005***

```

```

02-C — PRIMARY: TWO-WAY FIXED EFFECTS
02-C — PRIMARY: TWO-WAY FIXED EFFECTS

```

```

Estimator: PanelOLS(entity_effects=True, time_effects=True)

```

```
inst_avg 0.9000 0.4648 0.055 0.5690 0.116
inst_avg 0.9000 0.4648 0.055 0.5690 0.116
gdp_per_capita_log -2.0662 1.0288 0.047 0.8183 0.013 **
gdp_per_capita_log -2.0662 1.0288 0.047 0.8183 0.013 **
inflation -0.0507 0.0191 0.009 0.0222 0.024 **
inflation -0.0507 0.0191 0.009 0.0222 0.024 **
exports_percent_gdp 0.1038 0.0115 0.000 0.0300 0.001 ***
exports_percent_gdp 0.1038 0.0115 0.000 0.0300 0.001 ***
```

```
population_log 10.4267 4.0544 0.011 2.4373 0.000 ***
population_log 10.4267 4.0544 0.011 2.4373 0.000 ***
unemployment -0.1883 0.1404 0.182 0.1199 0.118
unemployment -0.1883 0.1404 0.182 0.1199 0.118
```

Note: Wild cluster bootstrap p-values in Module 04 (preferred for  $G=8$ ).

■ ■ EQ3: FDI + Institutions → GDP Growth ■ ■

EQ3:  $fdi + inst\_avg \rightarrow gdp\_growth$   
 EQ3:  $fdi + inst\_avg \rightarrow gdp\_growth$   
 N=179 | Within  $R^2=0.0949$  | DK bar  
 N=179 | Within  $R^2=0.0949$  | DK bar

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

1. **Identify the main topic of the text.**  
 2. **Summarize the main points of the text.**  
 3. **Identify the main topic of the text.**  
 4. **Summarize the main points of the text.**

```
fdi_percent_gdp 0.1371 0.1095 0.213 0.2063 0.508
fdi_percent_gdp 0.1371 0.1095 0.213 0.2063 0.508
inst_avg 0.5133 1.5841 0.746 1.1959 0.668
inst_avg 0.5133 1.5841 0.746 1.1959 0.668
gdp_per_capita_log 0.8887 1.2714 0.486 1.3187 0.501
gdp_per_capita_log 0.8887 1.2714 0.486 1.3187 0.501
inflation -0.1126 0.0291 0.000 0.0397 0.005 ***
inflation -0.1126 0.0291 0.000 0.0397 0.005 ***
exports_percent_gdp 0.0635 0.0382 0.099 0.0509 0.214
exports_percent_gdp 0.0635 0.0382 0.099 0.0509 0.214
population_log -0.7333 6.4156 0.909 5.4348 0.893
population_log -0.7333 6.4156 0.909 5.4348 0.893
unemployment -0.2101 0.1126 0.064 0.0887 0.019 **
unemployment -0.2101 0.1126 0.064 0.0887 0.019 **
```

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

\_\_\_\_\_

\_\_\_\_\_

02-D — METHODOLOGICAL NOTE: GENERATED REGRESSORS

\_\_\_\_\_

\_\_\_\_\_

- Sequential OLS and the generated regressors problem
- Sequential OLS and the generated regressors problem

The sequential estimation above uses observed values of `inst_avg` in EQ2 and observed `fdi` in EQ3. This is consistent (each equation is estimated independently) but does NOT recover the causal chain from a single identified system.

An alternative using fitted values ( $\text{inst\_avg\_hat}$  in EQ2,  $\text{fdi\_hat}$  in EQ3)  
 An alternative using fitted values ( $\text{inst\_avg\_hat}$  in EQ2,  $\text{fdi\_hat}$  in EQ3)  
 is a 'chain of OLS' approach, which suffers from the 'generated regressors'

is a 'chain of OLS' approach, which suffers from the 'generated regressors' problem (Pagan, 1984, Review of Economic Studies 51(2)):

problem (Pagan, 1984, Review of Economic Studies 51(2)):

- Standard errors in downstream equations are understated because they ignore sampling uncertainty in the upstream prediction.
- Standard errors in downstream equations are understated because they ignore sampling uncertainty in the upstream prediction.

Correct approaches:

Correct approaches:

- (a) 3SLS: estimate all three equations simultaneously.
- (a) 3SLS: estimate all three equations simultaneously.
- (b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.
- (b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.
- (c) IV/2SLS for each upstream equation with valid external instruments.
- (c) IV/2SLS for each upstream equation with valid external instruments.

In this pipeline:

In this pipeline:

- Module 02 reports observed-value FE (each equation self-contained).
- Module 02 reports observed-value FE (each equation self-contained).
- Module 04 bootstrap re-estimates the full system jointly, providing honest inference that integrates uncertainty across the chain.
- Module 04 bootstrap re-estimates the full system jointly, providing honest inference that integrates uncertainty across the chain.
- IV/2SLS is architecturally prepared but requires external instruments (cocaine price index, rainfall anomaly) not present in current dataset.
- IV/2SLS is architecturally prepared but requires external instruments (cocaine price index, rainfall anomaly) not present in current dataset.

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

Motivation: if FE and Between (cross-sectional average) estimates

Motivation: if FE and Between (cross-sectional average) estimates

diverge, this suggests omitted time-invariant heterogeneity —

diverge, this suggests omitted time-invariant heterogeneity —

supporting the use of Fixed Effects.

supporting the use of Fixed Effects.

Reference: Hausman (1978), Econometrica 46(6), 1251–1271.

Reference: Hausman (1978), Econometrica 46(6), 1251–1271.

EQ1: inst\_avg

EQ1: inst\_avg

Variable FE coef BE coef  $\Delta$

Variable FE coef BE coef  $\Delta$

homicide\_rate\_log\_lag1 -0.1404 0.0527 -0.1931■ ← divergence

homicide\_rate\_log\_lag1 -0.1404 0.0527 -0.1931■ ← divergence

gdp\_per\_capita\_log 0.1571 1.0182 -0.8611■ ← divergence

gdp\_per\_capita\_log 0.1571 1.0182 -0.8611■ ← divergence

inflation -0.0025 0.0807 -0.0832■ ← divergence

inflation -0.0025 0.0807 -0.0832■ ← divergence

exports\_percent\_gdp -0.0103 -0.0063 -0.0040 ■ ← divergence  
 exports\_percent\_gdp -0.0103 -0.0063 -0.0040 ■ ← divergence  
 population\_log -1.4015 -0.6008 -0.8007 ■ ← divergence  
 population\_log -1.4015 -0.6008 -0.8007 ■ ← divergence  
 unemployment 0.0099 0.1119 -0.1020 ■ ← divergence  
 unemployment 0.0099 0.1119 -0.1020 ■ ← divergence

EQ2: fdi\_percent\_gdp

EQ2: fdi\_percent\_gdp

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

inst\_avg 0.9000 0.0113 0.8887 ■ ← divergence  
 inst\_avg 0.9000 0.0113 0.8887 ■ ← divergence  
 gdp\_per\_capita\_log -2.0662 -0.1095 -1.9568 ■ ← divergence  
 gdp\_per\_capita\_log -2.0662 -0.1095 -1.9568 ■ ← divergence  
 inflation -0.0507 0.2575 -0.3082 ■ ← divergence  
 inflation -0.0507 0.2575 -0.3082 ■ ← divergence  
 exports\_percent\_gdp 0.1038 0.1309 -0.0272  
 exports\_percent\_gdp 0.1038 0.1309 -0.0272  
 population\_log 10.4267 -0.2008 10.6275 ■ ← divergence  
 population\_log 10.4267 -0.2008 10.6275 ■ ← divergence  
 unemployment -0.1883 0.4188 -0.6072 ■ ← divergence  
 unemployment -0.1883 0.4188 -0.6072 ■ ← divergence

EQ3: gdp\_growth

EQ3: gdp\_growth

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

fdi\_percent\_gdp 0.1371 0.1163 0.0208  
 fdi\_percent\_gdp 0.1371 0.1163 0.0208  
 inst\_avg 0.5133 -1.7183 2.2316 ■ ← divergence  
 inst\_avg 0.5133 -1.7183 2.2316 ■ ← divergence  
 gdp\_per\_capita\_log 0.8887 2.3911 -1.5024 ■ ← divergence  
 gdp\_per\_capita\_log 0.8887 2.3911 -1.5024 ■ ← divergence  
 inflation -0.1126 0.2678 -0.3804 ■ ← divergence  
 inflation -0.1126 0.2678 -0.3804 ■ ← divergence  
 exports\_percent\_gdp 0.0635 0.0196 0.0439 ■ ← divergence  
 exports\_percent\_gdp 0.0635 0.0196 0.0439 ■ ← divergence  
 population\_log -0.7333 -1.2294 0.4961  
 population\_log -0.7333 -1.2294 0.4961  
 unemployment -0.2101 0.1698 -0.3799 ■ ← divergence  
 unemployment -0.2101 0.1698 -0.3799 ■ ← divergence

02-F — RESIDUAL FIGURES

02-F — RESIDUAL FIGURES

✓ Figure saved → figures/04\_fe\_residuals.png

✓ Figure saved → figures/04\_fe\_residuals.png

02-G — EXPORT

02-G — EXPORT

- ✓ Dataset with predictions saved → panel\_with\_predictions.csv
- ✓ Dataset with predictions saved → panel\_with\_predictions.csv
- ✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\02\_fe\_results.json
- ✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\02\_fe\_results.json
- ✓ FE specifications saved → fe\_specs.pkl (for bootstrap module)
- ✓ FE specifications saved → fe\_specs.pkl (for bootstrap module)

Module 02 complete.

Module 02 complete.

✓ Module 02 completed in 11.0s

03-A — LOAD

03-A — LOAD

- ✓ Data loaded: 179 obs, 8 countries
- ✓ Data loaded: 179 obs, 8 countries

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section and Panel Data*, MIT Press, p.282-283.

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section and Panel Data*, MIT Press, p.282-283.

Procedure:

Procedure:

- (a) Estimate the model in FIRST DIFFERENCES:  $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$
- (a) Estimate the model in FIRST DIFFERENCES:  $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$
- (b) Regress  $\Delta \varepsilon_{it}$  on  $\Delta \varepsilon_{it-1}$  (OLS, pooled, no intercept)
- (b) Regress  $\Delta \varepsilon_{it}$  on  $\Delta \varepsilon_{it-1}$  (OLS, pooled, no intercept)
- (c) Test  $H_0: \rho = -0.5$
- (c) Test  $H_0: \rho = -0.5$

Under AR(1) errors in levels, the FD residuals have autocorrelation exactly -0.5.

Under AR(1) errors in levels, the FD residuals have autocorrelation exactly -0.5.

A t-test against -0.5 is the Wooldridge test.

A t-test against -0.5 is the Wooldridge test.

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test using robust standard errors and a slightly different form. The Python implementation below follows the same theoretical derivation but may differ numerically from Stata output.

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test using robust standard errors and a slightly different form. The Python implementation below follows the same theoretical derivation but may differ numerically from Stata output.



$T_{ij}$  is the number of common time periods.

Under  $H_0$  (no CD):  $CD \rightarrow N(0,1)$  as  $N, T \rightarrow \infty$ .

Under  $H_0$  (no CD):  $CD \rightarrow N(0,1)$  as  $N, T \rightarrow \infty$ .

Note: with  $N=8$ , the asymptotic distribution is approximate.

Note: with  $N=8$ , the asymptotic distribution is approximate.

eq1: inst\_avg

eq1: inst\_avg

CD = -66.7215,  $p = 0.0000$  \*\*\*

CD = -66.7215,  $p = 0.0000$  \*\*\*

N pairs = 28,  $T_{\text{min}} = 23$

N pairs = 28,  $T_{\text{min}} = 23$

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq2: fdi\_percent\_gdp

eq2: fdi\_percent\_gdp

CD = -64.7704,  $p = 0.0000$  \*\*\*

CD = -64.7704,  $p = 0.0000$  \*\*\*

N pairs = 28,  $T_{\text{min}} = 24$

N pairs = 28,  $T_{\text{min}} = 24$

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq3: gdp\_growth

eq3: gdp\_growth

CD = -72.1120,  $p = 0.0000$  \*\*\*

CD = -72.1120,  $p = 0.0000$  \*\*\*

N pairs = 28,  $T_{\text{min}} = 24$

N pairs = 28,  $T_{\text{min}} = 24$

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Tests  $H_0: \sigma^2_i = \sigma^2$  for all  $i$  (homoscedasticity across groups).

Tests  $H_0: \sigma^2_i = \sigma^2$  for all  $i$  (homoscedasticity across groups).

Statistic:  $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

Statistic:  $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

eq1: inst\_avg

eq1: inst\_avg

$W = 38.001$ ,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*

$W = 38.001$ ,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*

$\sigma^2_i$  by country:

$\sigma^2_i$  by country:

COL  $\sigma^2 = 0.0686$   $T=22.0$

COL  $\sigma^2 = 0.0686$   $T=22.0$

CRI  $\sigma^2 = 0.0196$   $T=22.0$

CRI  $\sigma^2 = 0.0196$   $T=22.0$

DOM  $\sigma^2 = 0.0368$   $T=21.0$

DOM  $\sigma^2=0.0368$   $T=21.0$   
 GTM  $\sigma^2=0.0096$   $T=20.0$   
 GTM  $\sigma^2=0.0096$   $T=20.0$   
 HND  $\sigma^2=0.0109$   $T=22.0$   
 HND  $\sigma^2=0.0109$   $T=22.0$   
 NIC  $\sigma^2=0.0598$   $T=19.0$   
 NIC  $\sigma^2=0.0598$   $T=19.0$   
 PAN  $\sigma^2=0.0133$   $T=22.0$   
 PAN  $\sigma^2=0.0133$   $T=22.0$   
 SLV  $\sigma^2=0.0362$   $T=23.0$   
 SLV  $\sigma^2=0.0362$   $T=23.0$   
 Pooled  $\sigma^2 = 0.0316$   
 Pooled  $\sigma^2 = 0.0316$   
 → ■ Heteroscedastic → clustered / DK SE already applied  
 → ■ Heteroscedastic → clustered / DK SE already applied

eq2: fdi\_percent\_gdp  
 eq2: fdi\_percent\_gdp  
 W = 67.638,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*  
 W = 67.638,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*  
 $\sigma^2_i$  by country:  
 $\sigma^2_i$  by country:  
 COL  $\sigma^2=1.0864$   $T=23.0$   
 COL  $\sigma^2=1.0864$   $T=23.0$   
 CRI  $\sigma^2=0.5696$   $T=23.0$   
 CRI  $\sigma^2=0.5696$   $T=23.0$   
 DOM  $\sigma^2=0.8312$   $T=22.0$   
 DOM  $\sigma^2=0.8312$   $T=22.0$

GTM  $\sigma^2=1.5648$   $T=21.0$   
 GTM  $\sigma^2=1.5648$   $T=21.0$   
 HND  $\sigma^2=0.9044$   $T=23.0$   
 HND  $\sigma^2=0.9044$   $T=23.0$   
 NIC  $\sigma^2=1.7567$   $T=20.0$   
 NIC  $\sigma^2=1.7567$   $T=20.0$   
 PAN  $\sigma^2=6.1034$   $T=23.0$   
 PAN  $\sigma^2=6.1034$   $T=23.0$   
 SLV  $\sigma^2=3.0253$   $T=24.0$   
 SLV  $\sigma^2=3.0253$   $T=24.0$   
 Pooled  $\sigma^2 = 2.0009$   
 Pooled  $\sigma^2 = 2.0009$   
 → ■ Heteroscedastic → clustered / DK SE already applied  
 → ■ Heteroscedastic → clustered / DK SE already applied

eq3: gdp\_growth  
 eq3: gdp\_growth  
 W = 40.945,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*  
 W = 40.945,  $\chi^2(7)$ ,  $p = 0.0000$  \*\*\*  
 $\sigma^2_i$  by country:  
 $\sigma^2_i$  by country:  
 COL  $\sigma^2=1.6686$   $T=23.0$   
 COL  $\sigma^2=1.6686$   $T=23.0$   
 CRI  $\sigma^2=1.6430$   $T=23.0$   
 CRI  $\sigma^2=1.6430$   $T=23.0$   
 DOM  $\sigma^2=4.3702$   $T=22.0$   
 DOM  $\sigma^2=4.3702$   $T=22.0$   
 GTM  $\sigma^2=3.4912$   $T=21.0$

GTM  $\sigma^2=3.4912$   $T=21.0$

HND  $\sigma^2=1.9061$   $T=23.0$

HND  $\sigma^2=1.9061$   $T=23.0$

NIC  $\sigma^2=3.9101$   $T=20.0$

NIC  $\sigma^2=3.9101$   $T=20.0$

PAN  $\sigma^2=8.1270$   $T=23.0$

PAN  $\sigma^2=8.1270$   $T=23.0$

SLV  $\sigma^2=1.0503$   $T=24.0$

SLV  $\sigma^2=1.0503$   $T=24.0$

Pooled  $\sigma^2 = 3.2391$

Pooled  $\sigma^2 = 3.2391$

→ ■ Heteroscedastic → clustered / DK SE already applied

→ ■ Heteroscedastic → clustered / DK SE already applied

TEST 4 — Multicollinearity: VIF and Condition Number

TEST 4 — Multicollinearity: VIF and Condition Number

VIF computed on within-transformed (two-way demeaned) regressors.

VIF computed on within-transformed (two-way demeaned) regressors.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

eq1: inst\_avg

eq1: inst\_avg

Variable VIF Flag

Variable VIF Flag

exports\_percent\_gdp 1.61 ✓ OK

exports\_percent\_gdp 1.61 ✓ OK

unemployment 1.48 ✓ OK

unemployment 1.48 ✓ OK

population\_log 1.43 ✓ OK

population\_log 1.43 ✓ OK

homicide\_rate\_log\_lag1 1.34 ✓ OK

homicide\_rate\_log\_lag1 1.34 ✓ OK

gdp\_per\_capita\_log 1.30 ✓ OK

gdp\_per\_capita\_log 1.30 ✓ OK

inflation 1.20 ✓ OK

inflation 1.20 ✓ OK

Condition number: 2.4 ✓ OK (<30)

Condition number: 2.4 ✓ OK (<30)

eq2: fdi\_percent\_gdp

eq2: fdi\_percent\_gdp

Variable VIF Flag

Variable VIF Flag

exports\_percent\_gdp 1.68 ✓ OK

exports\_percent\_gdp 1.68 ✓ OK

population\_log 1.42 ✓ OK

population\_log 1.42 ✓ OK

gdp\_per\_capita\_log 1.33 ✓ OK

gdp\_per\_capita\_log 1.33 ✓ OK  
 unemployment 1.22 ✓ OK  
 unemployment 1.22 ✓ OK  
 inst\_avg 1.19 ✓ OK  
 inst\_avg 1.19 ✓ OK  
 inflation 1.18 ✓ OK  
 inflation 1.18 ✓ OK  
 Condition number: 2.3 ✓ OK (<30)  
 Condition number: 2.3 ✓ OK (<30)

eq3: gdp\_growth  
 eq3: gdp\_growth  
 Variable VIF Flag  
 Variable VIF Flag

exports\_percent\_gdp 1.89 ✓ OK  
 exports\_percent\_gdp 1.89 ✓ OK  
 population\_log 1.49 ✓ OK  
 population\_log 1.49 ✓ OK  
 gdp\_per\_capita\_log 1.38 ✓ OK  
 gdp\_per\_capita\_log 1.38 ✓ OK  
 fdi\_percent\_gdp 1.32 ✓ OK  
 fdi\_percent\_gdp 1.32 ✓ OK  
 unemployment 1.26 ✓ OK  
 unemployment 1.26 ✓ OK  
 inst\_avg 1.21 ✓ OK  
 inst\_avg 1.21 ✓ OK  
 inflation 1.20 ✓ OK  
 inflation 1.20 ✓ OK

Condition number: 2.6 ✓ OK (<30)  
 Condition number: 2.6 ✓ OK (<30)

TEST 5 — Influence Diagnostics: DFBETA by Country  
 TEST 5 — Influence Diagnostics: DFBETA by Country

$$DFBETA_c = (\beta_{\text{full}} - \beta_{\text{full}_{-c}}) / SE(\beta_{\text{full}})$$

$$DFBETA_c = (\beta_{\text{full}} - \beta_{\text{full}_{-c}}) / SE(\beta_{\text{full}})$$
 where  $\beta_{\text{full}_{-c}}$  is the coefficient estimated excluding country c.  
 where  $\beta_{\text{full}_{-c}}$  is the coefficient estimated excluding country c.  
 $|DFBETA| \geq 1$  indicates influential observation.  
 $|DFBETA| \geq 1$  indicates influential observation.  
 Reference: Belsley, Kuh & Welsch (1980).  
 Reference: Belsley, Kuh & Welsch (1980).

eq1: inst\_avg | Key var: homicide\_rate\_log\_lag1  
 eq1: inst\_avg | Key var: homicide\_rate\_log\_lag1  
 $\beta_{\text{full}} = -0.1404$  SE = 0.1358  
 $\beta_{\text{full}} = -0.1404$  SE = 0.1358  
 Country  $\beta_{\text{full}_{-c}}$  DFBETA Flag  
 Country  $\beta_{\text{full}_{-c}}$  DFBETA Flag

eq2: fdi\_percent\_gdp | Key var: inst\_avg  
eq2: fdi\_percent\_gdp | Key var: inst\_avg  
 $\beta_{full} = 0.9000$  SE = 0.4648  
 $\beta_{full} = 0.9000$  SE = 0.4648  
Country  $\beta_{c}$  DFBETA Flag  
Country  $\beta_{c}$  DFBETA Flag

eq3: gdp\_growth | Key var: fdi\_percent\_gdp  
eq3: gdp\_growth | Key var: fdi\_percent\_gdp  
 $\beta_{full} = 0.1371$  SE = 0.1095  
 $\beta_{full} = 0.1371$  SE = 0.1095  
Country  $\beta_{\{c\}}$  DFBETA Flag  
Country  $\beta_{\{c\}}$  DFBETA Flag

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GTM 0.1572 -0.1838 ✓ OK  
GTM 0.1572 -0.1838 ✓ OK  
HND 0.1325 0.0418 ✓ OK  
HND 0.1325 0.0418 ✓ OK  
NIC 0.0754 0.5631 ✓ OK  
NIC 0.0754 0.5631 ✓ OK  
PAN 0.0077 1.1818 ✗ Influential (|DFBETA|>1)  
PAN 0.0077 1.1818 ✗ Influential (|DFBETA|>1)  
SLV 0.2082 -0.6493 ✓ OK  
SLV 0.2082 -0.6493 ✓ OK

03-F — DIAGNOSTIC FIGURE

03-F — DIAGNOSTIC FIGURE

✓ Figure saved → figures/05\_diagnostics.png

✓ Figure saved → figures/05\_diagnostics.png

03-G — REMEDIES SUMMARY

03-G — REMEDIES SUMMARY

Diagnostic Result Remedy applied

Diagnostic Result Remedy applied

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Wild bootstrap (Mod.04) ✓

Wild bootstrap (Mod.04) ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Group heterosced. See results above Clustered SE ✓

Group heterosced. See results above Clustered SE ✓

Multicollinearity See VIF above year\_c centring (Mod.01) ✓

Multicollinearity See VIF above year\_c centring (Mod.01) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\03\_diagnostics.json

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\03\_diagnostics.json

Module 03 complete.

Module 03 complete.

✓ Module 03 completed in 11.6s

04-A — LOAD

04-A — LOAD

✓ Data loaded. G=8 clusters, B=999 bootstrap draws.  
 ✓ Data loaded. G=8 clusters, B=999 bootstrap draws.  
 Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]  
 Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]  
 ■ With G=8, conventional clustered SE over-reject. Wild bootstrap preferred.  
 ■ With G=8, conventional clustered SE over-reject. Wild bootstrap preferred.

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)  
 04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

■ EQ1: dep=inst\_avg, key=homicide\_rate\_log\_lag1 ■  
 ■ EQ1: dep=inst\_avg, key=homicide\_rate\_log\_lag1 ■  
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration  
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration  
 Restricted model: PanelOLS without 'homicide\_rate\_log\_lag1' (H0:  $\beta=0$ )  
 Restricted model: PanelOLS without 'homicide\_rate\_log\_lag1' (H0:  $\beta=0$ )

b=200/999 elapsed=19s valid=200 fails=0  
 b=200/999 elapsed=19s valid=200 fails=0  
 b=400/999 elapsed=38s valid=400 fails=0  
 b=400/999 elapsed=38s valid=400 fails=0  
 b=600/999 elapsed=58s valid=600 fails=0  
 b=600/999 elapsed=58s valid=600 fails=0  
 b=800/999 elapsed=77s valid=800 fails=0  
 b=800/999 elapsed=77s valid=800 fails=0

Full-sample estimate:  
 Full-sample estimate:

$\beta = -0.1404$   
 $\beta = -0.1404$

SE (clustered, conventional) = 0.1358  
 SE (clustered, conventional) = 0.1358

t = -1.0336  
 t = -1.0336

Wild cluster bootstrap (Webb, B=999):  
 Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.4294  
 p-value (two-sided) = 0.4294

CI 95% (percentile) = [-0.2041, 0.2068]  
 CI 95% (percentile) = [-0.2041, 0.2068]

Valid draws: 999/999 | Fails: 0  
 Valid draws: 999/999 | Fails: 0

Runtime: 96.7s  
 Runtime: 96.7s

■  $\beta(\text{homicide\_rate\_log\_lag1})$  not significant (bootstrap p=0.4294)  
 ■  $\beta(\text{homicide\_rate\_log\_lag1})$  not significant (bootstrap p=0.4294)

Note: inspect sign and economic magnitude regardless of p-value.  
 Note: inspect sign and economic magnitude regardless of p-value.

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):  
 CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta = -0.1404$   
 $\beta = -0.1404$

SE(CR2) = 0.1420

```

SE(CR2) = 0.1420
t(CR2) = -0.9888
t(CR2) = -0.9888
df(Satterthwaite) = 3.16 [cf. G-1=7 used in v1 — different]
df(Satterthwaite) = 3.16 [cf. G-1=7 used in v1 — different]
p(CR2) = 0.3922
p(CR2) = 0.3922

■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
Restricted model: PanelOLS without 'inst_avg' (H0:  $\beta=0$ )
Restricted model: PanelOLS without 'inst_avg' (H0:  $\beta=0$ )
b=200/999 elapsed=17s valid=200 fails=0
b=200/999 elapsed=17s valid=200 fails=0
b=400/999 elapsed=35s valid=400 fails=0
b=400/999 elapsed=35s valid=400 fails=0

b=600/999 elapsed=54s valid=600 fails=0
b=600/999 elapsed=54s valid=600 fails=0
b=800/999 elapsed=73s valid=800 fails=0
b=800/999 elapsed=73s valid=800 fails=0

Full-sample estimate:
Full-sample estimate:
 $\beta$  ■ = 0.9000
 $\beta$  ■ = 0.9000
SE (clustered, conventional) = 0.4648
SE (clustered, conventional) = 0.4648
t = 1.9363
t = 1.9363

Wild cluster bootstrap (Webb, B=999):
Wild cluster bootstrap (Webb, B=999):
p-value (two-sided) = 0.0621 *
p-value (two-sided) = 0.0621 *
CI 95% (percentile) = [-0.8424, 0.8392]
CI 95% (percentile) = [-0.8424, 0.8392]
Valid draws: 999/999 | Fails: 0
Valid draws: 999/999 | Fails: 0
Runtime: 92.6s
Runtime: 92.6s
✓  $\beta$ (inst_avg) significant at 10% (bootstrap p=0.0621)
✓  $\beta$ (inst_avg) significant at 10% (bootstrap p=0.0621)

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 $\beta$  ■ = 0.9000
 $\beta$  ■ = 0.9000
SE(CR2) = 0.5155
SE(CR2) = 0.5155
t(CR2) = 1.7460
t(CR2) = 1.7460
df(Satterthwaite) = 2.38 [cf. G-1=7 used in v1 — different]
df(Satterthwaite) = 2.38 [cf. G-1=7 used in v1 — different]
p(CR2) = 0.2023
p(CR2) = 0.2023

```

■■ EQ3: dep=gdp\_growth, key=fdi\_percent\_gdp ■■

■■ EQ3: dep=gdp\_growth, key=fdi\_percent\_gdp ■■

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

Restricted model: PanelOLS without 'fdi\_percent\_gdp' (H0:  $\beta=0$ )

Restricted model: PanelOLS without 'fdi\_percent\_gdp' (H0:  $\beta=0$ )

b=200/999 elapsed=20s valid=200 fails=0

b=200/999 elapsed=20s valid=200 fails=0

b=400/999 elapsed=40s valid=400 fails=0

b=400/999 elapsed=40s valid=400 fails=0

b=600/999 elapsed=58s valid=600 fails=0

b=600/999 elapsed=58s valid=600 fails=0

b=800/999 elapsed=77s valid=800 fails=0

b=800/999 elapsed=77s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

$\beta = 0.1371$

$\beta = 0.1371$

SE (clustered, conventional) = 0.1095

SE (clustered, conventional) = 0.1095

t = 1.2522

t = 1.2522

Wild cluster bootstrap (Webb, B=999):

Wild cluster bootstrap (Webb, B=999):

p-value (two-sided) = 0.4094

p-value (two-sided) = 0.4094

CI 95% (percentile) = [-0.2404, 0.2412]

CI 95% (percentile) = [-0.2404, 0.2412]

Valid draws: 999/999 | Fails: 0

Valid draws: 999/999 | Fails: 0

Runtime: 95.1s

Runtime: 95.1s

■  $\beta(\text{fdi\_percent\_gdp})$  not significant (bootstrap p=0.4094)

■  $\beta(\text{fdi\_percent\_gdp})$  not significant (bootstrap p=0.4094)

Note: inspect sign and economic magnitude regardless of p-value.

Note: inspect sign and economic magnitude regardless of p-value.

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta = 0.1371$

$\beta = 0.1371$

SE(CR2) = 0.1133

SE(CR2) = 0.1133

t(CR2) = 1.2095

t(CR2) = 1.2095

df(Satterthwaite) = 1.79 [cf. G-1=7 used in v1 — different]

df(Satterthwaite) = 1.79 [cf. G-1=7 used in v1 — different]

p(CR2) = 0.3618

p(CR2) = 0.3618

04-C — SE COMPARISON TABLE

04-C — SE COMPARISON TABLE

Equation Dep. var. Key var.  $\beta$  ■ SE(Cl.) p(Cl.) SE(DK) SE(CR2) df\_satt p(WCB)

Equation Dep. var. Key var.  $\beta$  ■ SE(Cl.) p(Cl.) SE(DK) SE(CR2) df\_satt p(WCB)

eq1 inst\_avg homicide\_rate\_log\_lag1 -0.1404 0.1358 0.303 0.1282 0.1420 3.16 0.429

eq1 inst\_avg homicide\_rate\_log\_lag1 -0.1404 0.1358 0.303 0.1282 0.1420 3.16 0.429

eq2 fdi\_percent\_gdp inst\_avg 0.9000 0.4648 0.055 0.5690 0.5155 2.38 0.062\*

eq2 fdi\_percent\_gdp inst\_avg 0.9000 0.4648 0.055 0.5690 0.5155 2.38 0.062\*

eq3 gdp\_growth fdi\_percent\_gdp 0.1371 0.1095 nan nan 0.1133 1.79 0.409

eq3 gdp\_growth fdi\_percent\_gdp 0.1371 0.1095 nan nan 0.1133 1.79 0.409

Columns:

Columns:

SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

SE(CR2) with df\_Satterthwaite as secondary check.

SE(CR2) with df\_Satterthwaite as secondary check.

04-D — BOOTSTRAP FIGURES

04-D — BOOTSTRAP FIGURES

✓ Figure saved → figures/06\_bootstrap\_distributions.png

✓ Figure saved → figures/06\_bootstrap\_distributions.png

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\04\_bootstrap.json

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\04\_bootstrap.json

Module 04 complete.

Module 04 complete.

✓ Module 04 completed in 292.8s

05-A — LOAD

05-A — LOAD

✓ Data: 179 obs | 8 countries

✓ Data: 179 obs | 8 countries

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

## ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)



Bias addressed: results driven by a single influential country.

Bias addressed: results driven by a single influential country.

EQ1: *inst\_avg*

EQ1: *inst\_avg*

Specification N  $\beta$  SE p Stars  $\Delta\beta$

Specification N  $\beta$  SE p Stars  $\Delta\beta$



Full sample (baseline) 171 -0.1404 0.1358 0.303 +0.0000

Full sample (baseline) 171 -0.1404 0.1358 0.303 +0.0000

Excl. COL 149 -0.0197 0.1164 0.866 +0.1207

Excl. COL 149 -0.0197 0.1164 0.866 +0.1207

Excl. CRI 149 -0.2547 0.1383 0.068 \* -0.1143

Excl. CRI 149 -0.2547 0.1383 0.068 \* -0.1143

Excl. DOM 150 -0.0990 0.1569 0.529 +0.0414

Excl. DOM 150 -0.0990 0.1569 0.529 +0.0414

Excl. GTM 151 -0.1523 0.1396 0.277 -0.0119

Excl. GTM 151 -0.1523 0.1396 0.277 -0.0119

Excl. HND 149 -0.1554 0.1490 0.299 -0.0150

Excl. HND 149 -0.1554 0.1490 0.299 -0.0150

Excl. NIC 152 -0.1277 0.0930 0.172 +0.0127

Excl. NIC 152 -0.1277 0.0930 0.172 +0.0127

Excl. PAN 149 -0.1087 0.1401 0.439 +0.0317

Excl. PAN 149 -0.1087 0.1401 0.439 +0.0317

Excl. SLV 148 -0.2668 0.1241 0.034 \*\* -0.1264

Excl. SLV 148 -0.2668 0.1241 0.034 \*\* -0.1264

■ 0 sign reversal(s), 3 large shift(s) — check influential countries.

■ 0 sign reversal(s), 3 large shift(s) — check influential countries.

EQ2: *fdi\_percent\_gdp*

EQ2: *fdi\_percent\_gdp*

Specification N  $\beta$  SE p Stars  $\Delta\beta$

Specification N  $\beta$  SE p Stars  $\Delta\beta$



Full sample (baseline) 179 0.9000 0.4648 0.055 \* +0.0000

Full sample (baseline) 179 0.9000 0.4648 0.055 \* +0.0000

Excl. COL 156 1.7060 0.7673 0.028 \*\* +0.8060

Excl. COL 156 1.7060 0.7673 0.028 \*\* +0.8060

Excl. CRI 156 0.7640 0.5851 0.194 -0.1360

Excl. CRI 156 0.7640 0.5851 0.194 -0.1360

Excl. DOM 157 0.6559 0.3659 0.076 \* -0.2441

Excl. DOM 157 0.6559 0.3659 0.076 \* -0.2441

Excl. GTM 158 0.8322 0.2540 0.001 \*\*\* -0.0678

Excl. GTM 158 0.8322 0.2540 0.001 \*\*\* -0.0678

Excl. HND 156 0.7092 0.5282 0.182 -0.1908

Excl. HND 156 0.7092 0.5282 0.182 -0.1908

Excl. NIC 159 0.7702 0.6706 0.253 -0.1298

Excl. NIC 159 0.7702 0.6706 0.253 -0.1298  
 Excl. PAN 156 0.8802 0.4812 0.070 \* -0.0198  
 Excl. PAN 156 0.8802 0.4812 0.070 \* -0.0198  
 Excl. SLV 155 0.6992 0.7003 0.320 -0.2008  
 Excl. SLV 155 0.6992 0.7003 0.320 -0.2008  
 ■ 0 sign reversal(s), 1 large shift(s) — check influential countries.  
 ■ 0 sign reversal(s), 1 large shift(s) — check influential countries.

EQ3: gdp\_growth

EQ3: gdp\_growth

Specification N β ■ SE p Stars Δβ

Specification N β ■ SE p Stars Δβ

Full sample (baseline) 179 0.1371 0.1095 0.213 +0.0000

Full sample (baseline) 179 0.1371 0.1095 0.213 +0.0000

Excl. COL 156 0.1020 0.1043 0.330 -0.0351

Excl. COL 156 0.1020 0.1043 0.330 -0.0351

Excl. CRI 156 0.1376 0.1112 0.218 +0.0005

Excl. CRI 156 0.1376 0.1112 0.218 +0.0005

Excl. DOM 157 0.2015 0.0956 0.037 \*\* +0.0644

Excl. DOM 157 0.2015 0.0956 0.037 \*\* +0.0644

Excl. GTM 158 0.1572 0.1020 0.126 +0.0201

Excl. GTM 158 0.1572 0.1020 0.126 +0.0201

Excl. HND 156 0.1325 0.1230 0.284 -0.0046

Excl. HND 156 0.1325 0.1230 0.284 -0.0046

Excl. NIC 159 0.0754 0.1275 0.555 -0.0616

Excl. NIC 159 0.0754 0.1275 0.555 -0.0616

Excl. PAN 156 0.0077 0.1494 0.959 -0.1294

Excl. PAN 156 0.0077 0.1494 0.959 -0.1294

Excl. SLV 155 0.2082 0.1280 0.106 +0.0711

Excl. SLV 155 0.2082 0.1280 0.106 +0.0711

■ 0 sign reversal(s), 2 large shift(s) — check influential countries.

■ 0 sign reversal(s), 2 large shift(s) — check influential countries.

ROBUSTNESS 2 — Time Windows

ROBUSTNESS 2 — Time Windows

Bias addressed: period-specific shocks, structural breaks.

Bias addressed: period-specific shocks, structural breaks.

EQ1: inst\_avg

EQ1: inst\_avg

Specification N β ■ SE p Stars Δβ

Specification N β ■ SE p Stars Δβ

Full sample 171 -0.1404 0.1358 0.303 +0.0000

Full sample 171 -0.1404 0.1358 0.303 +0.0000

Pre-2010 (pre-GFC) 64 -0.1277 0.1443 0.381 +0.0127

Pre-2010 (pre-GFC) 64 -0.1277 0.1443 0.381 +0.0127  
 Post-2010 107 0.0252 0.0660 0.704 +0.1656  
 Post-2010 107 0.0252 0.0660 0.704 +0.1656  
 Excl. COVID (2020–21) 156 -0.1042 0.1188 0.382 +0.0362  
 Excl. COVID (2020–21) 156 -0.1042 0.1188 0.382 +0.0362  
 Post-peace (2014+) 75 0.0797 0.1053 0.453 +0.2201  
 Post-peace (2014+) 75 0.0797 0.1053 0.453 +0.2201

EQ2: fdi\_percent\_gdp

EQ2: fdi\_percent\_gdp

Specification N  $\beta$  SE p Stars  $\Delta\beta$

Specification N  $\beta$  SE p Stars  $\Delta\beta$



Full sample 179 0.9000 0.4648 0.055 \* +0.0000  
 Full sample 179 0.9000 0.4648 0.055 \* +0.0000  
 Pre-2010 (pre-GFC) 72 -0.0124 0.8363 0.988 -0.9124  
 Pre-2010 (pre-GFC) 72 -0.0124 0.8363 0.988 -0.9124  
 Post-2010 107 1.9317 0.9407 0.043 \*\* +1.0317  
 Post-2010 107 1.9317 0.9407 0.043 \*\* +1.0317  
 Excl. COVID (2020–21) 164 1.0179 0.3159 0.002 \*\*\* +0.1179  
 Excl. COVID (2020–21) 164 1.0179 0.3159 0.002 \*\*\* +0.1179  
 Post-peace (2014+) 75 3.6099 0.7574 0.000 \*\*\* +2.7099  
 Post-peace (2014+) 75 3.6099 0.7574 0.000 \*\*\* +2.7099

EQ3: gdp\_growth

EQ3: gdp\_growth

Specification N  $\beta$  SE p Stars  $\Delta\beta$

Specification N  $\beta$  SE p Stars  $\Delta\beta$



Full sample 179 0.1371 0.1095 0.213 +0.0000  
 Full sample 179 0.1371 0.1095 0.213 +0.0000  
 Pre-2010 (pre-GFC) 72 0.0512 0.1244 0.682 -0.0859  
 Pre-2010 (pre-GFC) 72 0.0512 0.1244 0.682 -0.0859  
 Post-2010 107 0.2680 0.1771 0.134 +0.1309  
 Post-2010 107 0.2680 0.1771 0.134 +0.1309  
 Excl. COVID (2020–21) 164 -0.0079 0.0945 0.933 -0.1450  
 Excl. COVID (2020–21) 164 -0.0079 0.0945 0.933 -0.1450  
 Post-peace (2014+) 75 0.0842 0.2595 0.747 -0.0529  
 Post-peace (2014+) 75 0.0842 0.2595 0.747 -0.0529



ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)

ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)



Uses pre-computed lag columns from dataset (not re-computed).

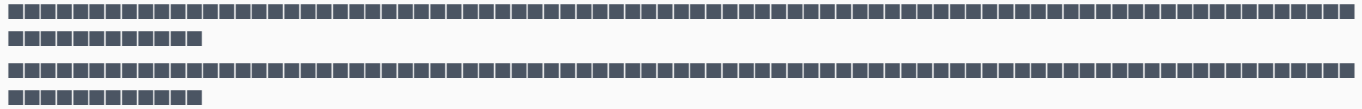
Uses pre-computed lag columns from dataset (not re-computed).

Bias addressed: uncertainty about transmission delay of violence.

Bias addressed: uncertainty about transmission delay of violence.

EQ1: Alternative Lags

## EQ1: Alternative Lags

Specification N  $\beta$  SE p Stars  $\Delta\beta$ Specification N  $\beta$  SE p Stars  $\Delta\beta$ 

Contemporaneous (lag 0) 179 -0.0512 0.1080 0.636 +0.0892

Contemporaneous (lag 0) 179 -0.0512 0.1080 0.636 +0.0892

Lag 1 (primary) 171 -0.1404 0.1358 0.303 +0.0000

Lag 1 (primary) 171 -0.1404 0.1358 0.303 +0.0000

Lag 2 163 -0.1930 0.1670 0.250 -0.0526

Lag 2 163 -0.1930 0.1670 0.250 -0.0526

Lag 3 155 -0.1836 0.1813 0.313 -0.0432

Lag 3 155 -0.1836 0.1813 0.313 -0.0432



## ROBUSTNESS 4 — Alternative Institution Indices

ROBUSTNESS 4 — Alternative Institution Indices



Bias addressed: sensitivity to index construction methodology.

Bias addressed: sensitivity to index construction methodology.

## EQ1: Alternative Institution Indices

EQ1: Alternative Institution Indices

Specification N  $\beta$  SE p Stars  $\Delta\beta$ Specification N  $\beta$  SE p Stars  $\Delta\beta$ 

inst\_avg (primary) 171 -0.1404 0.1358 0.303 +0.0000

inst\_avg (primary) 171 -0.1404 0.1358 0.303 +0.0000

inst\_pca (PCA-based) 171 -0.2333 0.2264 0.305 -0.0929

inst\_pca (PCA-based) 171 -0.2333 0.2264 0.305 -0.0929

rule\_of\_law 171 -1.8345 1.5064 0.225 -1.6941

rule\_of\_law 171 -1.8345 1.5064 0.225 -1.6941

control\_corruption 171 0.8270 0.9159 0.368 +0.9674

control\_corruption 171 0.8270 0.9159 0.368 +0.9674

political\_stability 171 -3.3434 3.1976 0.298 -3.2030

political\_stability 171 -3.3434 3.1976 0.298 -3.2030

■ Mixed signs across institution specifications — investigate.

■ Mixed signs across institution specifications — investigate.



## ROBUSTNESS 5 — Alternative Control Sets

ROBUSTNESS 5 — Alternative Control Sets



Bias addressed: over-controlling or omitted variable bias.

Bias addressed: over-controlling or omitted variable bias.

EQ1: inst\_avg

EQ1: inst\_avg

Baseline controls 171 -0.1404 0.1358 0.303 +0.0000  
 Baseline controls 171 -0.1404 0.1358 0.303 +0.0000  
 Extended controls — — — — FAILED  
 Extended controls — — — — FAILED  
 Minimal controls 171 -0.1034 0.1475 0.484 +0.0370  
 Minimal controls 171 -0.1034 0.1475 0.484 +0.0370  
 No controls (FE only) 171 -0.0825 0.1702 0.629 +0.0579  
 No controls (FE only) 171 -0.0825 0.1702 0.629 +0.0579

Baseline controls 179 0.9000 0.4648 0.055 \* +0.0000  
Baseline controls 179 0.9000 0.4648 0.055 \* +0.0000  
Extended controls — — — — FAILED  
Extended controls — — — — FAILED  
Minimal controls 179 0.5491 0.8413 0.515 -0.3509  
Minimal controls 179 0.5491 0.8413 0.515 -0.3509  
No controls (FE only) 179 -0.1192 0.8270 0.886 -1.0192  
No controls (FE only) 179 -0.1192 0.8270 0.886 -1.0192

Baseline controls 179 0.1450 0.1212 0.234 +0.0000  
 Baseline controls 179 0.1450 0.1212 0.234 +0.0000  
 Extended controls — — — FAILED  
 Extended controls — — — FAILED  
 Minimal controls 179 0.2666 0.1352 0.050 \* +0.1216  
 Minimal controls 179 0.2666 0.1352 0.050 \* +0.1216  
 No controls (FE only) 179 0.2355 0.1354 0.084 \* +0.0905  
 No controls (FE only) 179 0.2355 0.1354 0.084 \* +0.0905

Procedure: randomly permute homicide\_rate\_log\_lag1 within year cells.  
 Procedure: randomly permute homicide\_rate\_log\_lag1 within year cells.

This breaks the country-specific causal link while preserving the cross-sectional and temporal distribution of the variable.

*This breaks the country-specific causal link while preserving the cross-sectional and temporal distribution of the variable.*

*H0 (placebo): permuted violence has no effect on institutions.*

*H0 (placebo): permuted violence has no effect on institutions.*

*If the TRUE estimate is more extreme than 95% of placebo estimates,*

*If the TRUE estimate is more extreme than 95% of placebo estimates,*

*this confirms the result is not driven by spurious correlation.*

*this confirms the result is not driven by spurious correlation.*

*N\_permutations = 500.*

*N\_permutations = 500.*

True estimate:  $\beta_{\text{■}} = -0.1404$

True estimate:  $\beta_{\text{■}} = -0.1404$

Placebo distribution: mean=0.0006, SD=0.0243, 5th pctl=-0.0404

Placebo distribution: mean=0.0006, SD=0.0243, 5th pctl=-0.0404

Placebo p-value: 0.0000

Placebo p-value: 0.0000

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

05-G — FIGURES

05-G — FIGURES

✓ Figure saved → figures/07\_robustness.png

✓ Figure saved → figures/07\_robustness.png

05-H — EXPORT

05-H — EXPORT

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\05\_robustness.json

✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\05\_robustness.json

Module 05 complete.

Module 05 complete.

✓ Module 05 completed in 65.6s

06-A — LOAD

06-A — LOAD

✓ Data: 179 obs | 8 countries

✓ Data: 179 obs | 8 countries

CV strategy: Leave-One-Country-Out (8 folds)

CV strategy: Leave-One-Country-Out (8 folds)

■ All ML results are EXPLORATORY. They do not imply causation.

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■ G=8 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

■ G=8 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

■ T1: Violence → Institutions ■

■ T1: Violence → Institutions ■

Missing summary:

Missing summary:

*homicide\_rate\_log\_lag2* 0.089385

*homicide\_rate\_log\_lag2* 0.089385

*homicide\_rate\_log\_lag1* 0.044693

*homicide\_rate\_log\_lag1* 0.044693

*inst\_avg* 0.000000

*inst\_avg* 0.000000

*gdp\_per\_capita\_log* 0.000000

*gdp\_per\_capita\_log* 0.000000

*inflation* 0.000000

*inflation* 0.000000

*unemployment* 0.000000

*unemployment* 0.000000

*exports\_percent\_gdp* 0.000000

*exports\_percent\_gdp* 0.000000

*population\_log* 0.000000

*population\_log* 0.000000

*year\_c* 0.000000

*year\_c* 0.000000

*dtype: float64*

*dtype: float64*

Rows before dropna: 179

Rows before dropna: 179

Rows after dropna: 163

Rows after dropna: 163

Model  $R^2$  mean  $R^2$  SD RMSE mean Valid folds

Model  $R^2$  mean  $R^2$  SD RMSE mean Valid folds

Random Forest -33.1112 49.1064 0.7235 8

Random Forest -33.1112 49.1064 0.7235 8

Gradient Boosting -49.0544 74.8629 0.8054 8

Gradient Boosting -49.0544 74.8629 0.8054 8

■ Note: G=8 LOCO folds →  $R^2$  estimates have wide uncertainty. SD across folds is as informative as the mean.

■ Note: G=8 LOCO folds →  $R^2$  estimates have wide uncertainty. SD across folds is as informative as the mean.

RF vs GB Wilcoxon signed-rank (paired LOCO  $R^2$ ):

RF vs GB Wilcoxon signed-rank (paired LOCO  $R^2$ ):

N pairs: 8 | W=16.0 p=0.84375

N pairs: 8 | W=16.0 p=0.84375

■ G=8 pairs → very low power. Treat p-value as descriptive only.

■ G=8 pairs → very low power. Treat p-value as descriptive only.

RF OOB  $R^2$  (internal diagnostic — in-sample, optimistic): 0.8181

RF OOB  $R^2$  (internal diagnostic — in-sample, optimistic): 0.8181

■ OOB  $R^2$  is a training-data statistic, NOT a LOCO out-of-country  $R^2$ . Do not cite OOB  $R^2$  as evidence of predictive performance.

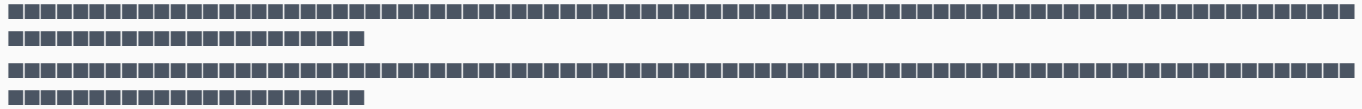
■ OOB  $R^2$  is a training-data statistic, NOT a LOCO out-of-country  $R^2$ . Do not cite OOB  $R^2$  as evidence of predictive performance.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)



Population (log) 0.3159 0.3473±0.0335 0.2440 1.61 0.93 0.96 0.99

Population (log) 0.3159 0.3473±0.0335 0.2440 1.61 0.93 0.96 0.99

GDP per capita (log) 0.2029 0.1752±0.0227 0.1676 2.52 1.42 0.73 0.94

GDP per capita (log) 0.2029 0.1752±0.0227 0.1676 2.52 1.42 0.73 0.94

Violence lag 1 0.1314 0.0721±0.0106 0.1077 4.34 1.26 0.23 0.84

Violence lag 1 0.1314 0.0721±0.0106 0.1077 4.34 1.26 0.23 0.84

Violence lag 2 0.1264 0.0703±0.0097 0.1020 3.17 1.52 0.70 0.90

Violence lag 2 0.1264 0.0703±0.0097 0.1020 3.17 1.52 0.70 0.90

Unemployment 0.0789 0.0576±0.0093 0.0650 5.34 1.43 0.10 0.57

Unemployment 0.0789 0.0576±0.0093 0.0650 5.34 1.43 0.10 0.57

Exports (% GDP) 0.0780 0.0492±0.0045 0.0429 5.03 1.55 0.17 0.51

Exports (% GDP) 0.0780 0.0492±0.0045 0.0429 5.03 1.55 0.17 0.51

Inflation 0.0476 0.0323±0.0041 0.0341 6.28 1.38 0.10 0.23

Inflation 0.0476 0.0323±0.0041 0.0341 6.28 1.38 0.10 0.23

Time Trend (centred) 0.0188 0.0119±0.0019 0.0188 7.71 0.65 0.00 0.02

Time Trend (centred) 0.0188 0.0119±0.0019 0.0188 7.71 0.65 0.00 0.02

Key feature 'homicide\_rate\_log\_lag1':

Key feature 'homicide\_rate\_log\_lag1':

Permutation rank: #3/8

Permutation rank: #3/8

SHAP rank: #3/8

SHAP rank: #3/8

✓ Key feature ranks in top 3 by both importance metrics.

✓ Key feature ranks in top 3 by both importance metrics.

■■ T2: Institutions → FDI ■■

■■ T2: Institutions → FDI ■■

Missing summary:

Missing summary:

tourist\_arrivals\_log 0.128492

tourist\_arrivals\_log 0.128492

homicide\_rate\_log\_lag1 0.044693

homicide\_rate\_log\_lag1 0.044693

fdi\_percent\_gdp 0.000000

fdi\_percent\_gdp 0.000000

gdp\_per\_capita\_log 0.000000

gdp\_per\_capita\_log 0.000000

inst\_avg 0.000000

inst\_avg 0.000000

inflation 0.000000

inflation 0.000000

unemployment 0.000000

unemployment 0.000000

exports\_percent\_gdp 0.000000

exports\_percent\_gdp 0.000000

year\_c 0.000000

year\_c 0.000000

Key feature 'inst\_avg':  
Key feature 'inst\_avg':  
Permutation rank: #4/8

■ OOB  $R^2$  is a training-data statistic, NOT a LOCO out-of-country  $R^2$ . Do not cite OOB  $R^2$  as evidence of predictive performance.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

FDI (% GDP) 0.1833 0.1102±0.0147 0.3284 1.98 1.13 0.91 0.98

FDI (% GDP) 0.1833 0.1102±0.0147 0.3284 1.98 1.13 0.91 0.98

Time Trend (centred) 0.1988 0.1760±0.0195 0.3029 4.17 1.65 0.42 0.80

Time Trend (centred) 0.1988 0.1760±0.0195 0.3029 4.17 1.65 0.42 0.80

Exports (% GDP) 0.1438 0.0743±0.0125 0.3021 3.02 2.00 0.62 0.86

Exports (% GDP) 0.1438 0.0743±0.0125 0.3021 3.02 2.00 0.62 0.86

Inflation 0.1300 0.0677±0.0123 0.2243 4.30 1.89 0.34 0.69

Inflation 0.1300 0.0677±0.0123 0.2243 4.30 1.89 0.34 0.69

GDP per capita (log) 0.0818 0.0407±0.0073 0.2009 5.20 1.70 0.18 0.55

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Unemployment 0.1173 0.0540±0.0074 0.1391 5.58 1.86 0.16 0.44

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Violence lag 1 0.0702 0.0315±0.0060 0.1357 6.16 2.19 0.17 0.28

Violence lag 1 0.0702 0.0315±0.0060 0.1357 6.16 2.19 0.17 0.28

Institution Index (avg) 0.0748 0.0260±0.0037 0.0917 5.58 2.23 0.19 0.41

Institution Index (avg) 0.0748 0.0260±0.0037 0.0917 5.58 2.23 0.19 0.41

Key feature 'fdi\_percent\_gdp':

Key feature 'fdi\_percent\_gdp':

Permutation rank: #2/8

Permutation rank: #2/8

SHAP rank: #1/8

SHAP rank: #1/8

✓ Key feature ranks in top 3 by both importance metrics.

✓ Key feature ranks in top 3 by both importance metrics.

06-C — CONVERGENCE TABLE: FE vs ML

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Convergence principle (Athey & Imbens 2019):

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If the variables that matter causally (significant in FE) also rank

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highly in ML importance, this triangulation strengthens the narrative.

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Divergence signals potential non-linearities or confounders.

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Task FE coef FE p-val FE sig Direction Perm rank SHAP rank Convergence

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T1: Violence → Institutions ? 0.3032 3 3 ✓ Strong

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T2: Institutions → FDI ? 0.0548 \* 4 3 ■ Partial  
 T2: Institutions → FDI ? 0.0548 \* 4 3 ■ Partial  
 T3: FDI + Institutions → Growth ? ? ? 2 1 ✓ Strong  
 T3: FDI + Institutions → Growth ? ? ? 2 1 ✓ Strong

Notes:

Notes:

- FE coef and p-value are cluster-robust estimates from Module 02.
- FE coef and p-value are cluster-robust estimates from Module 02.

- FE sig: \* p<0.10 \*\* p<0.05 \*\*\* p<0.01.
- FE sig: \* p<0.10 \*\* p<0.05 \*\*\* p<0.01.
- Direction: sign of the FE coefficient for the key feature.
- Direction: sign of the FE coefficient for the key feature.
- Perm/SHAP rank: rank among all features in the ML model (1 = most important).
- Perm/SHAP rank: rank among all features in the ML model (1 = most important).
- Convergence: Strong = key feature ranks top 3 by both metrics.
- Convergence: Strong = key feature ranks top 3 by both metrics.

06-D — FIGURES

06-D — FIGURES

- ✓ Figure saved → figures/08\_ml\_loco\_cv.png
- ✓ Figure saved → figures/08\_ml\_loco\_cv.png
- ✓ Figure saved → figures/09\_feature\_importance.png
- ✓ Figure saved → figures/09\_feature\_importance.png
- ✓ Figure saved → figures/10\_shap\_ale.png
- ✓ Figure saved → figures/10\_shap\_ale.png

06-E — EXPORT

06-E — EXPORT

- ✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\06\_ml\_results.json
- ✓ Saved → C:\data\_analyzer\econometric\_pipeline\pipeline\json\06\_ml\_results.json

Module 06 complete. Pipeline finished.

Module 06 complete. Pipeline finished.

- IMPORTANT REMINDER ■
- IMPORTANT REMINDER ■
- ML results (SHAP, permutation importance, LOCO R<sup>2</sup>) ■
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- are EXPLORATORY and NON-CAUSAL. ■
- are EXPLORATORY and NON-CAUSAL. ■
- They triangulate — they do not identify causal effects. ■
- They triangulate — they do not identify causal effects. ■
- Primary causal evidence: Module 02 (Two-Way FE) and ■
- Primary causal evidence: Module 02 (Two-Way FE) and ■
- Module 04 (Wild Cluster Bootstrap inference). ■
- Module 04 (Wild Cluster Bootstrap inference). ■
- ■

- ■
- SHAP limitation: computed on the full-sample model. ■
- SHAP limitation: computed on the full-sample model. ■
- Values reflect in-sample feature associations and may not ■
- Values reflect in-sample feature associations and may not ■

- generalise to held-out countries. ■
- generalise to held-out countries. ■



- Bootstrap stability limitation:  $G=8$  clusters  $\rightarrow$  discrete ■
- Bootstrap stability limitation:  $G=8$  clusters  $\rightarrow$  discrete ■
- bootstrap distribution. Treat rank SDs and hit rates as ■
- bootstrap distribution. Treat rank SDs and hit rates as ■
- approximate lower bounds on true uncertainty. ■
- approximate lower bounds on true uncertainty. ■

✓ Module 06 completed in 1922.9s

## PIPELINE SUMMARY

- ✓ Module 01: Data Preparation & Institution Index
- ✓ Module 02: Panel Estimation (Two-Way FE)
- ✓ Module 03: Panel Diagnostics
- ✓ Module 04: Wild Cluster Bootstrap Inference
- ✓ Module 05: Robustness Checks
- ✓ Module 06: ML Triangulation

Total runtime: 2326.0s

*Output directories:*

Figures → C:\data\_analyzer\econometric\_pipeline\pipeline\figures\

Tables → C:\data\_analyzer\econometric\_pipeline\pipeline\tables\

JSON → C:\data\_analyzer\econometric\_pipeline\pipeline\json\